



(12) **United States Patent**
Cheng et al.

(10) **Patent No.:** **US 12,402,407 B2**
(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **GAP-INSULATED SEMICONDUCTOR DEVICE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/672,218**

(22) Filed: **May 23, 2024**

(65) **Prior Publication Data**
US 2024/0312994 A1 Sep. 19, 2024

Related U.S. Application Data
(63) Continuation of application No. 18/079,691, filed on Dec. 12, 2022, now Pat. No. 11,996,410, which is a (Continued)

(51) **Int. Cl.**
H01L 27/092 (2006.01)
H01L 21/8238 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 84/85** (2025.01); **H10D 30/031** (2025.01); **H10D 30/6715** (2025.01);
(Continued)

(58) **Field of Classification Search**
CPC H01L 27/092; H01L 21/823807; H01L 21/823814; H01L 21/823878;
(Continued)

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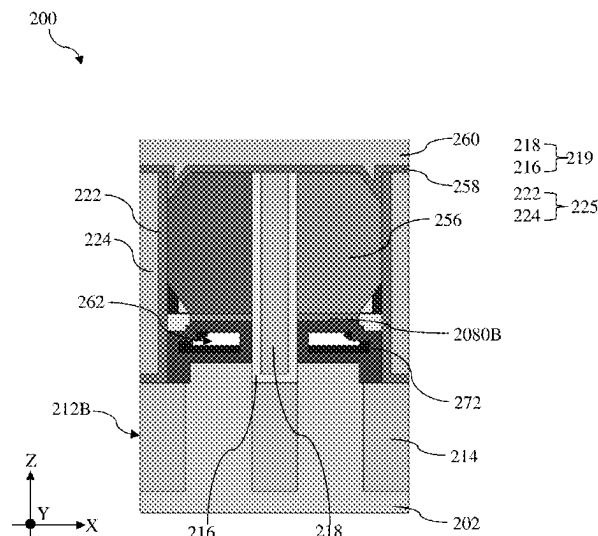
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(57) **ABSTRACT**

Semiconductor structures and the manufacturing method thereof are disclosed. An exemplary semiconductor structure according to the present disclosure includes a first base portion and a second base portion, an isolation feature sandwiched between the first base portion and the second base portion, a center dielectric fin over the isolation feature, a first anti-punch-through (APT) feature over the first base portion, a second APT feature over the second base portion, a first stack of channel members over the first APT feature, and a second stack of channel members over the second APT feature. The center dielectric fin is sandwiched between the first stack of channel members and the second stack of channel members as well as between the first APT feature and the second APT feature.

20 Claims, 28 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/142,640, filed on
Jan. 6, 2021, now Pat. No. 11,527,534.

(51) **Int. Cl.**

H01L 29/06 (2006.01)
H01L 29/423 (2006.01)
H01L 29/66 (2006.01)
H01L 29/786 (2006.01)
H10D 30/01 (2025.01)
H10D 30/67 (2025.01)
H10D 62/10 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)
H10D 84/85 (2025.01)

(52) **U.S. Cl.**

CPC *H10D 30/6735* (2025.01); *H10D 30/6757*
(2025.01); *H10D 62/118* (2025.01); *H10D*
84/0167 (2025.01); *H10D 84/017* (2025.01);
H10D 84/0188 (2025.01); *H10D 84/038*
(2025.01)

(58) **Field of Classification Search**

CPC H01L 29/0665; H01L 29/42392; H01L
29/66742; H01L 29/78621; H01L
29/78696; H01L 29/1079; H01L

21/823481; H01L 29/0649; H01L
29/0673; H01L 29/66439; H01L 29/775;
H01L 27/088; H01L 27/0886; H01L
21/823431; B82Y 10/00

USPC 257/288

See application file for complete search history.

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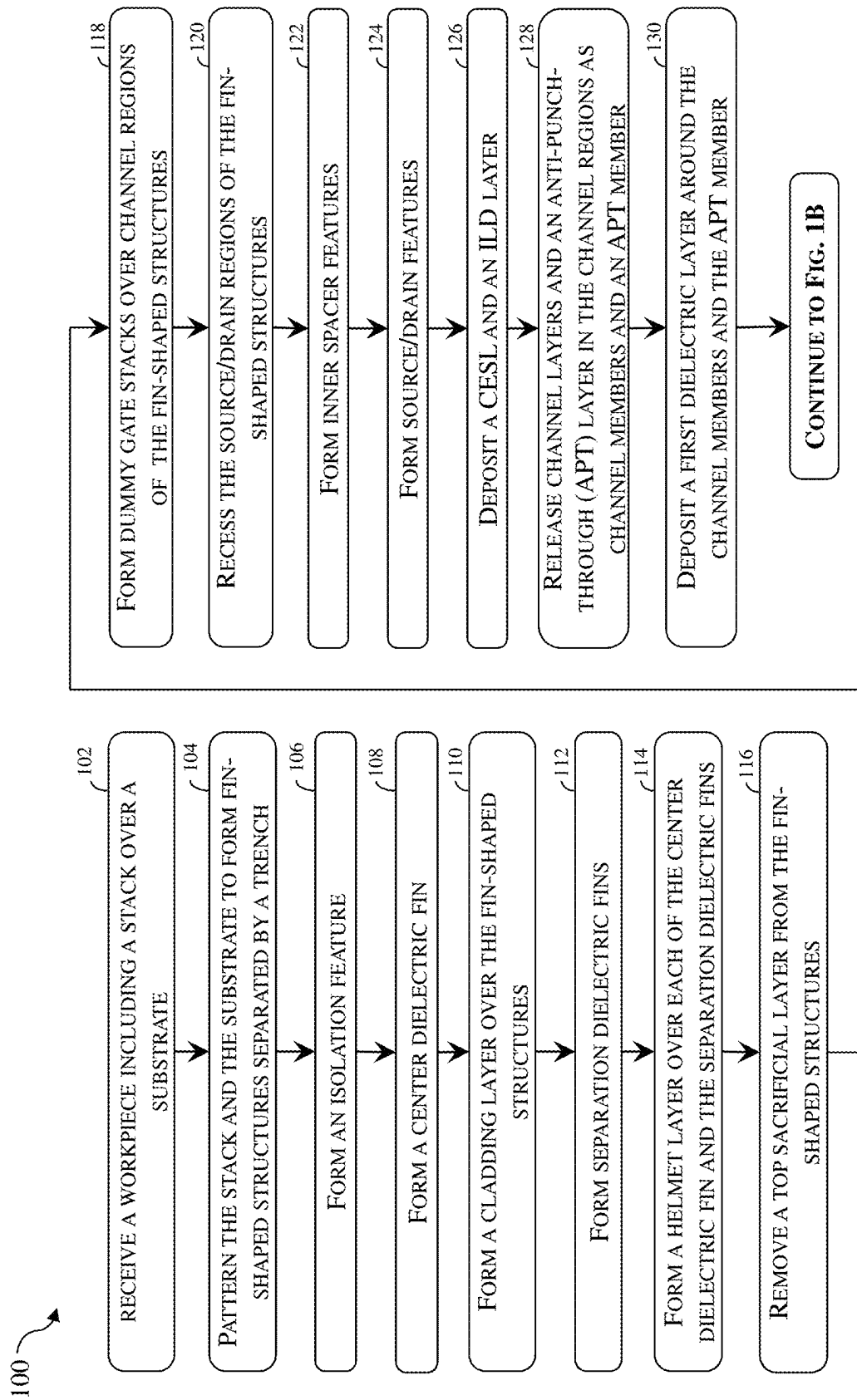


Fig. 1A

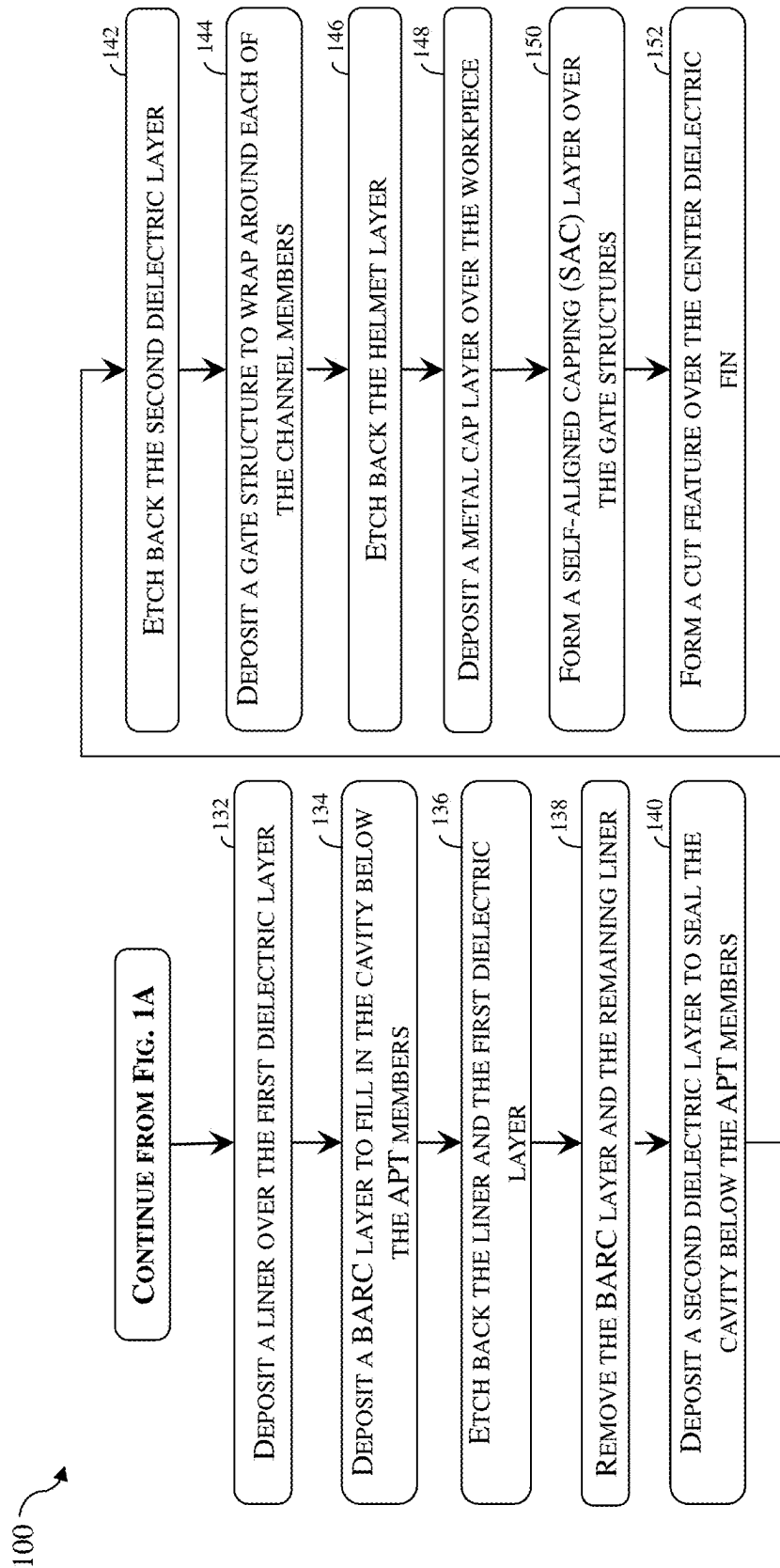


Fig. 1B

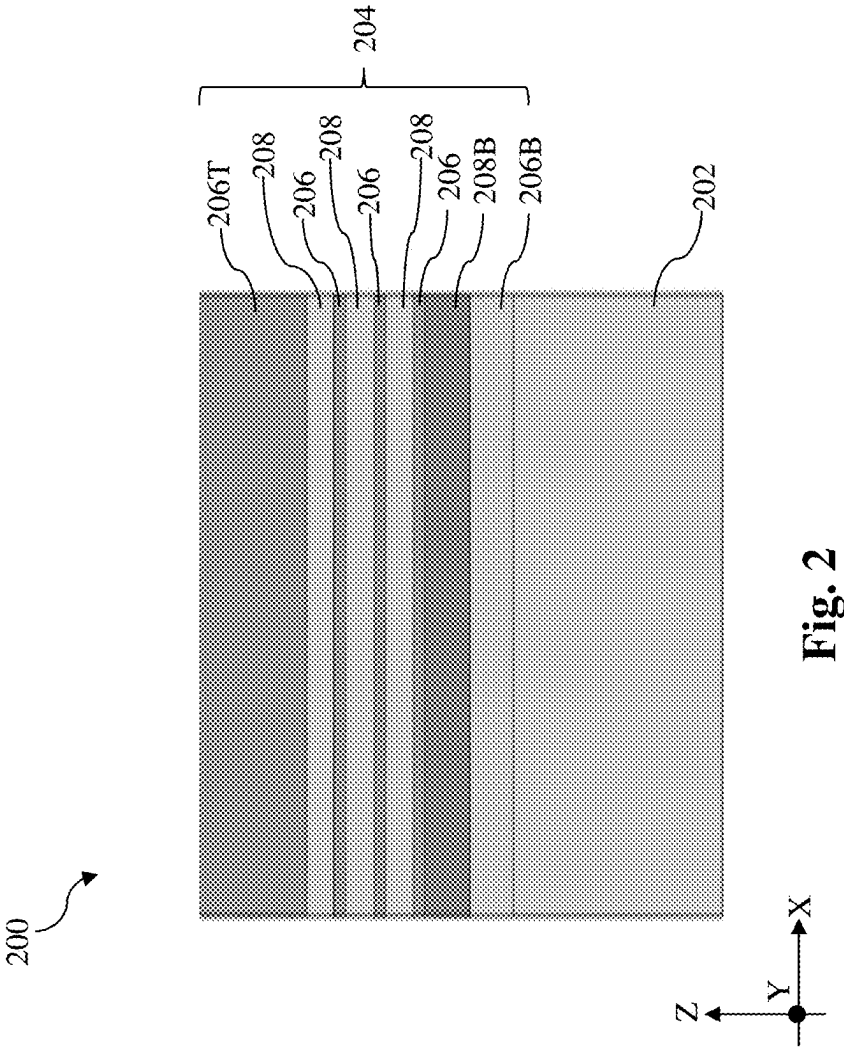
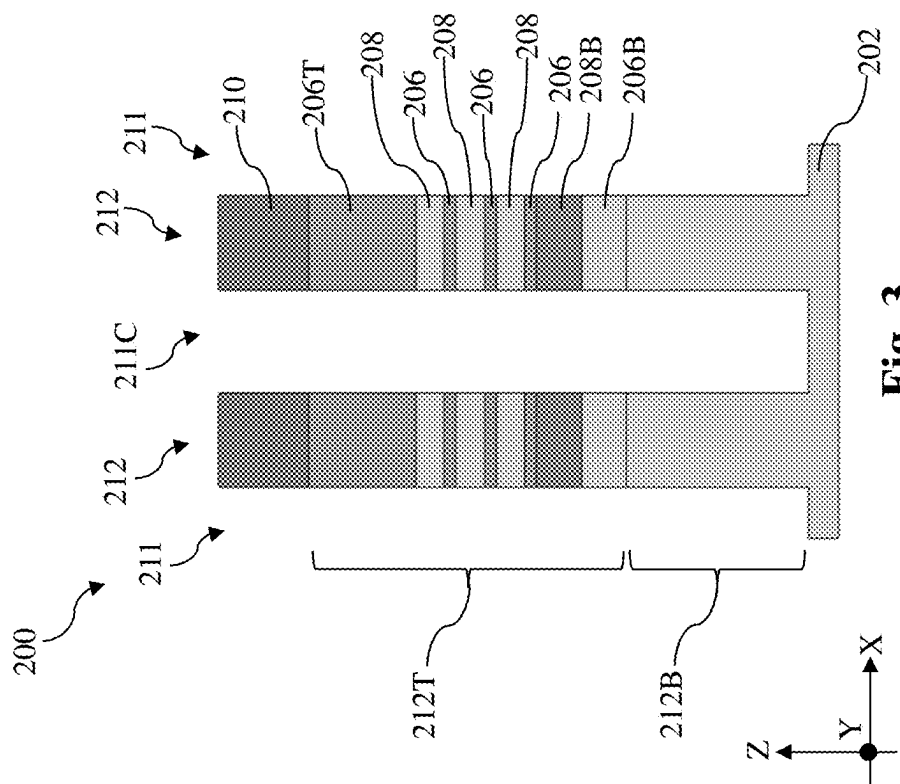
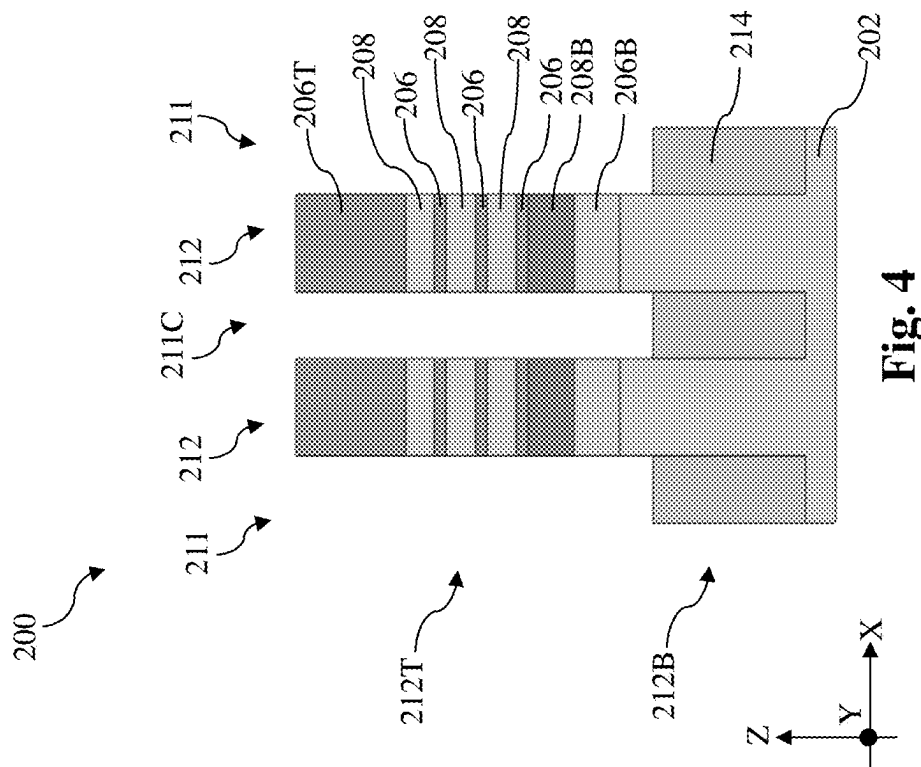
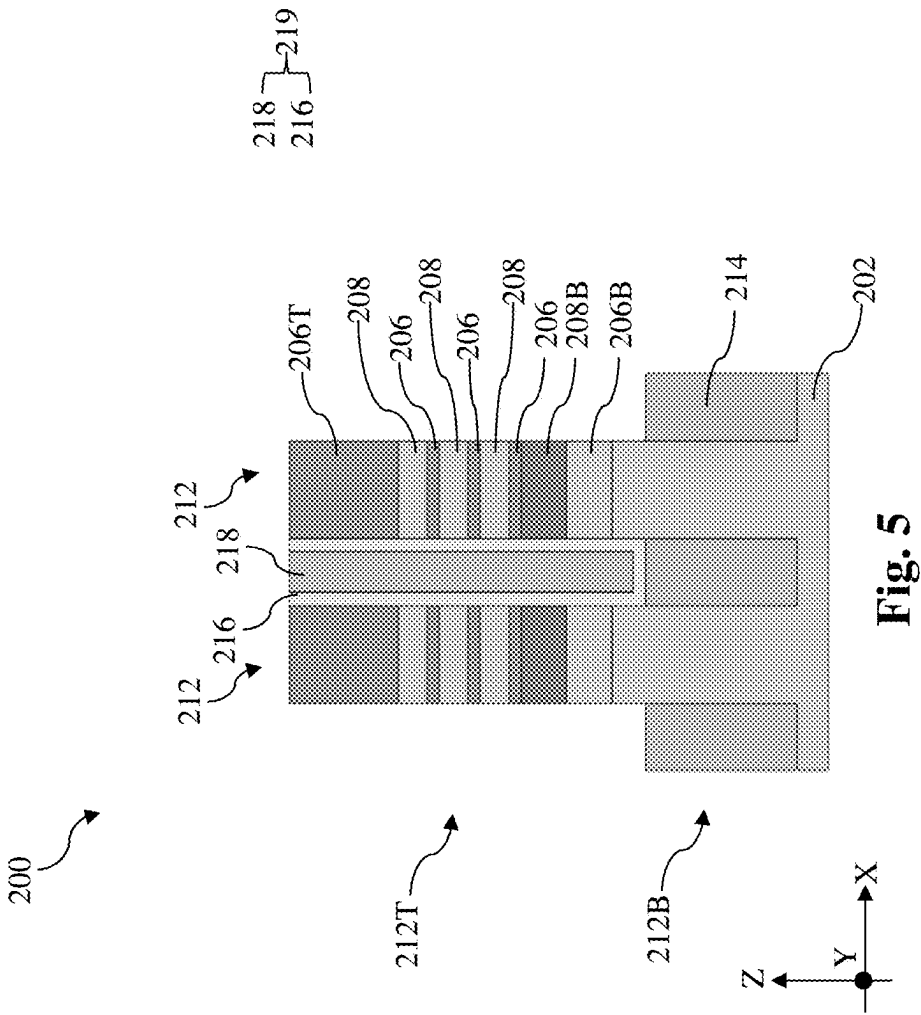
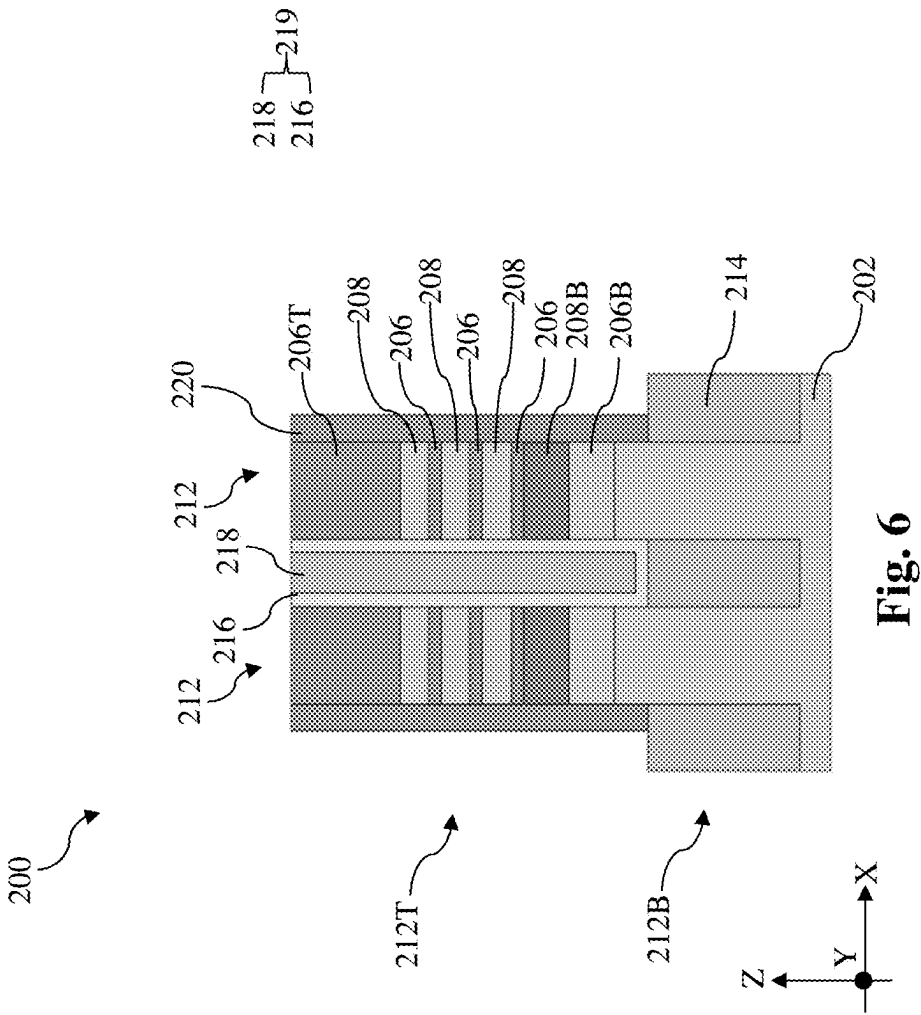


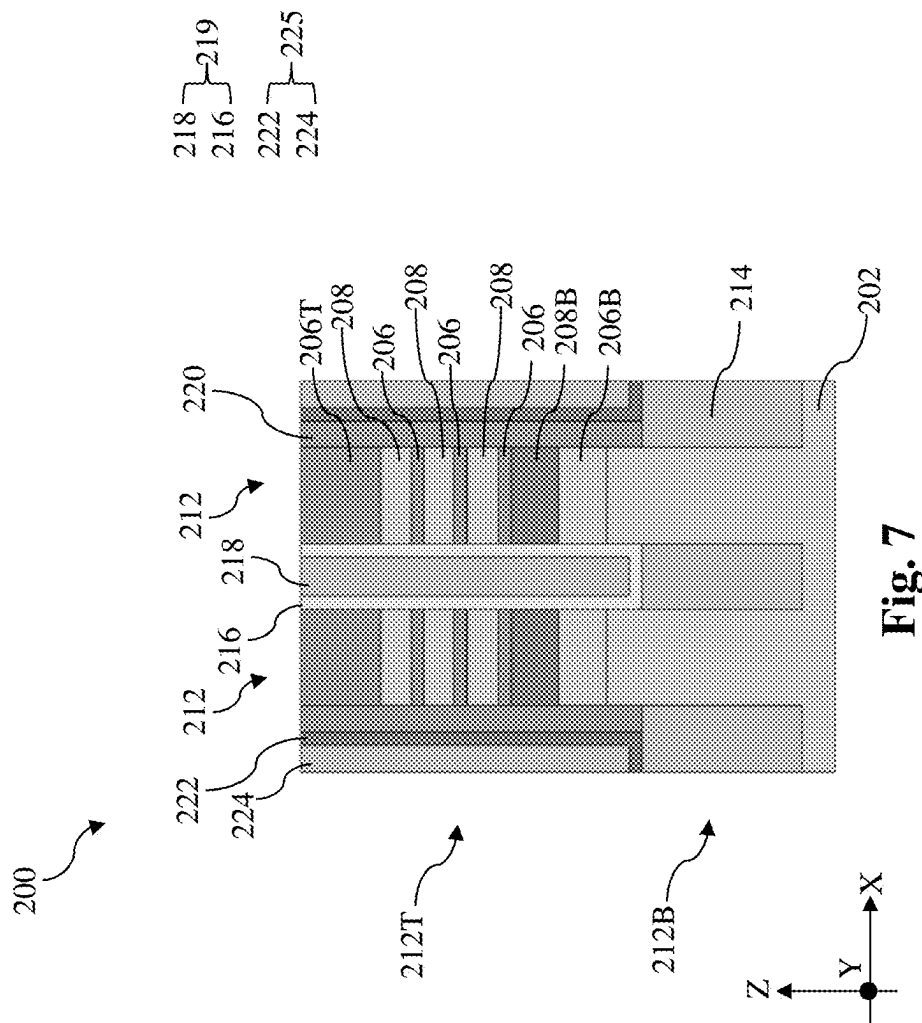
Fig. 2

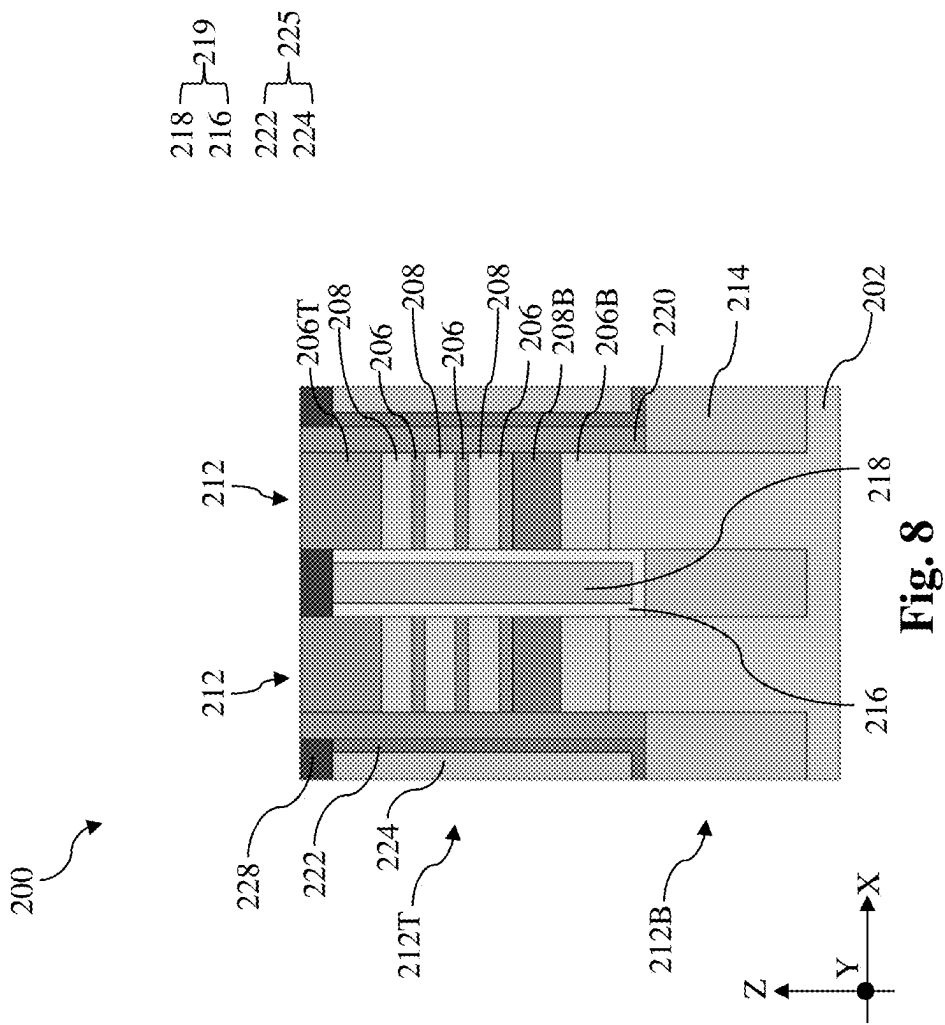


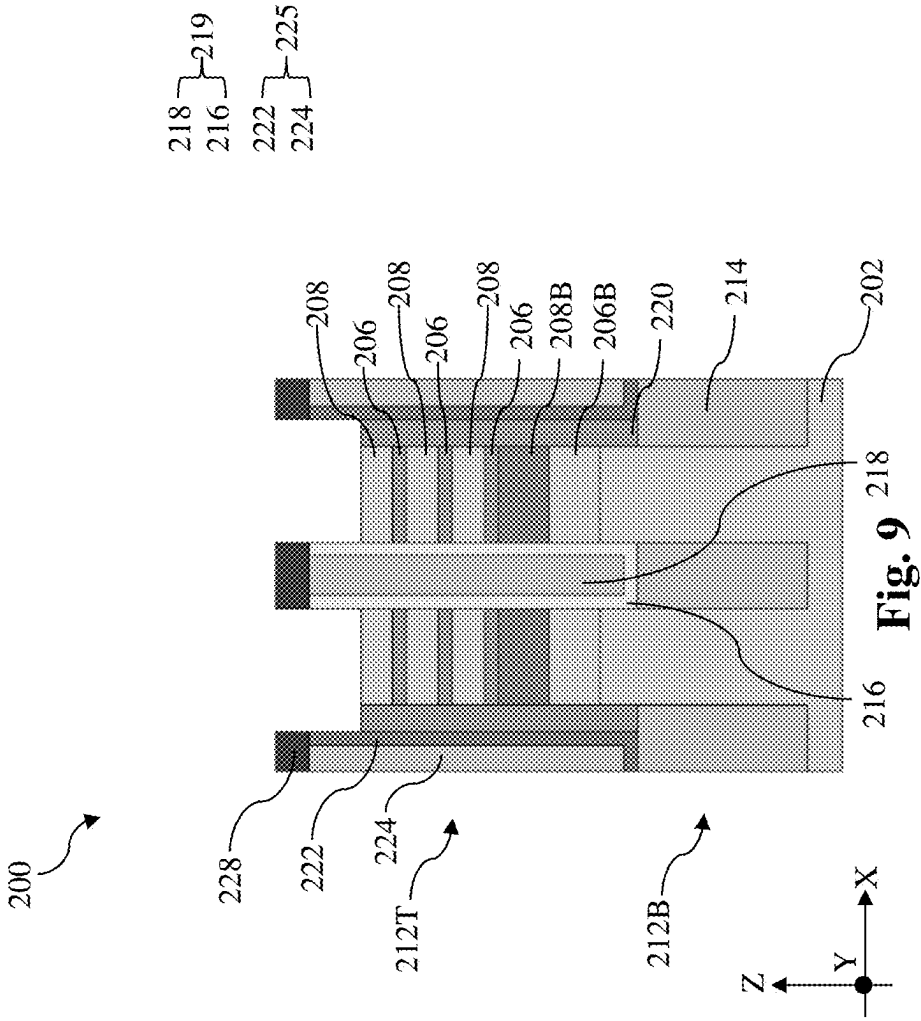


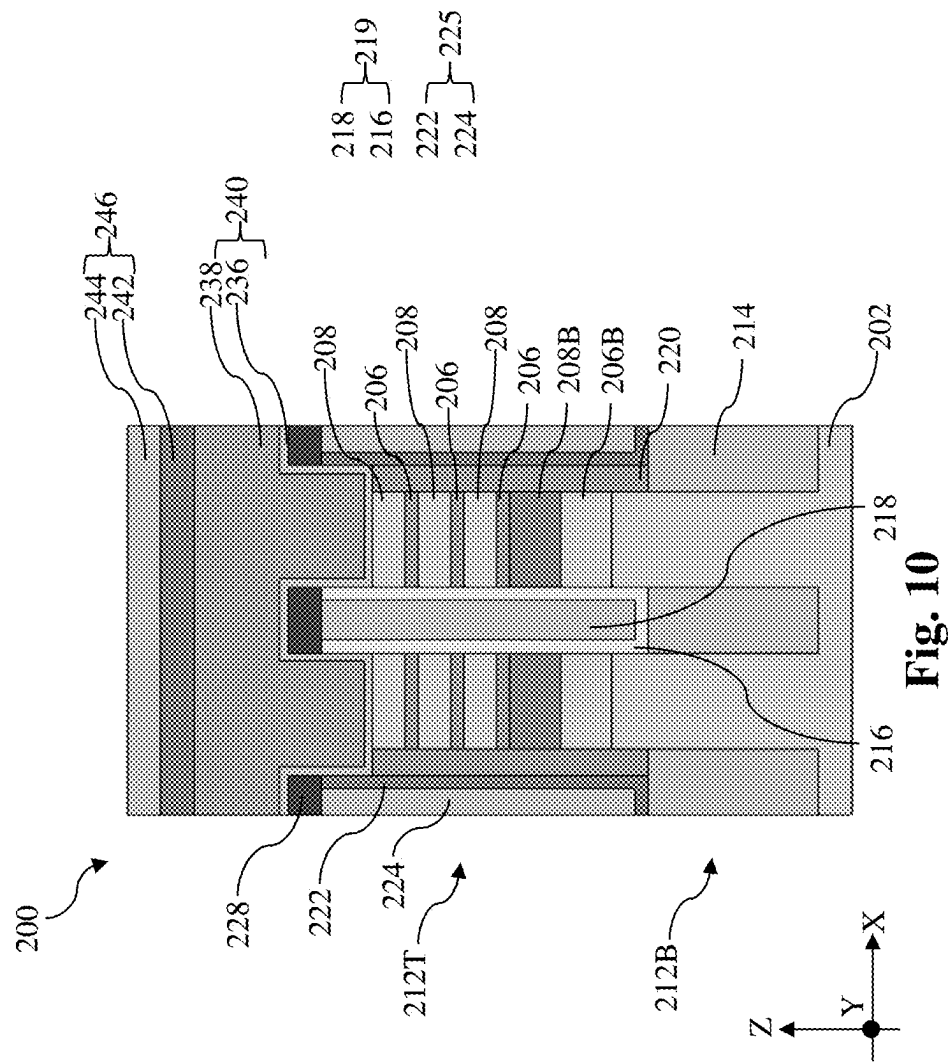












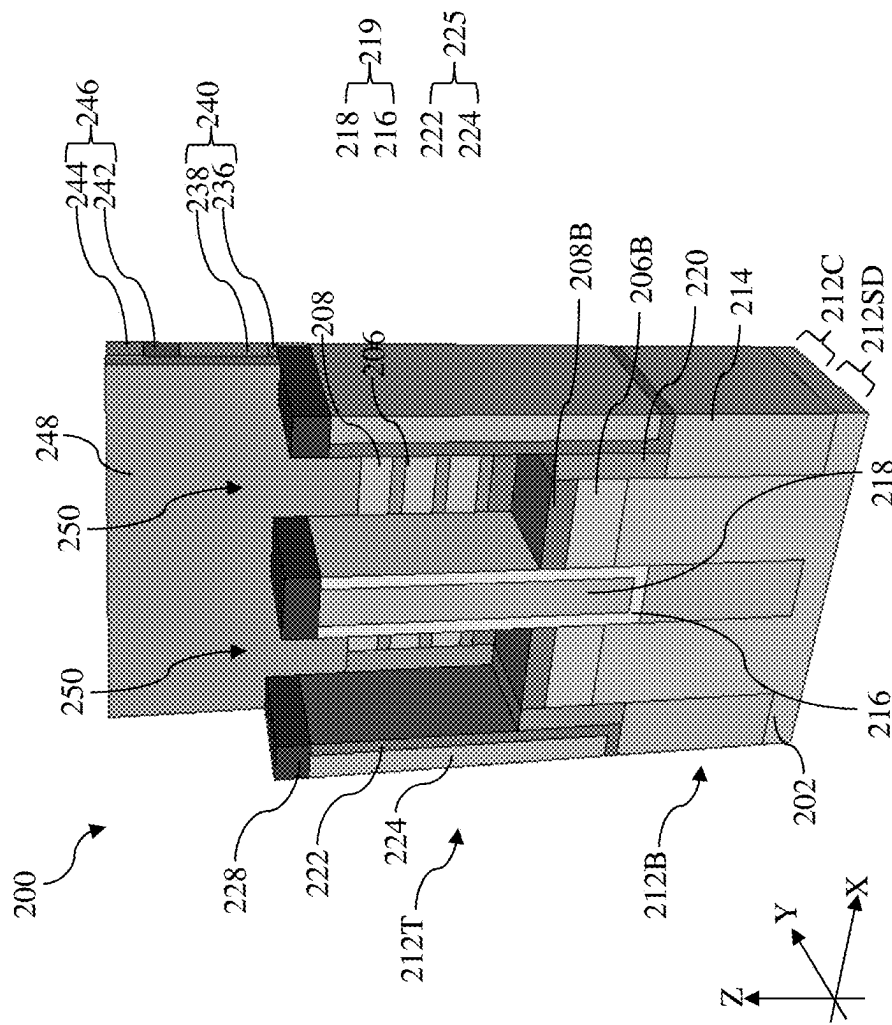


Fig. 11

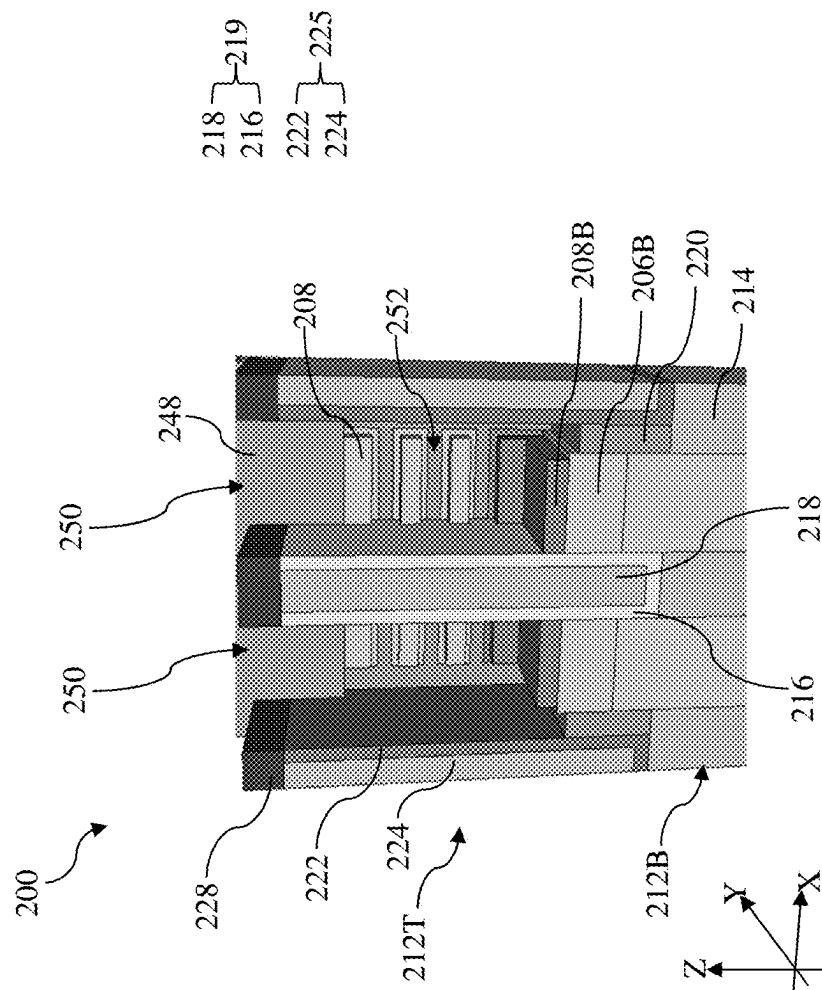


Fig. 12

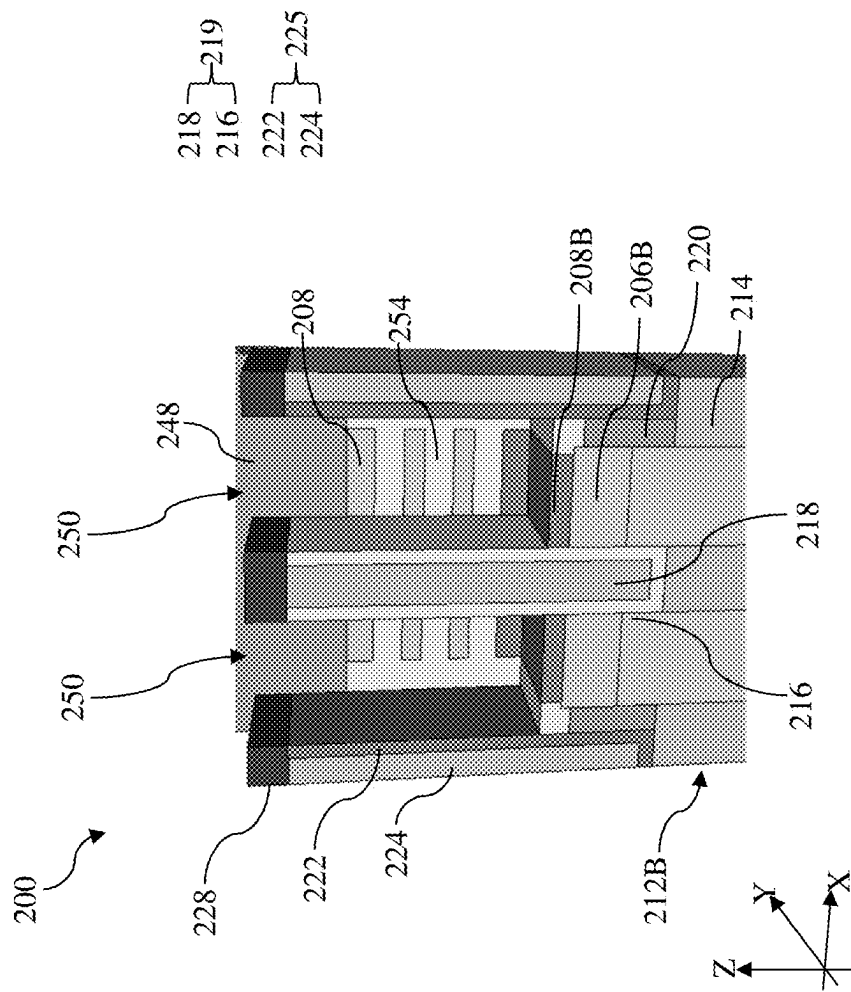


Fig. 13

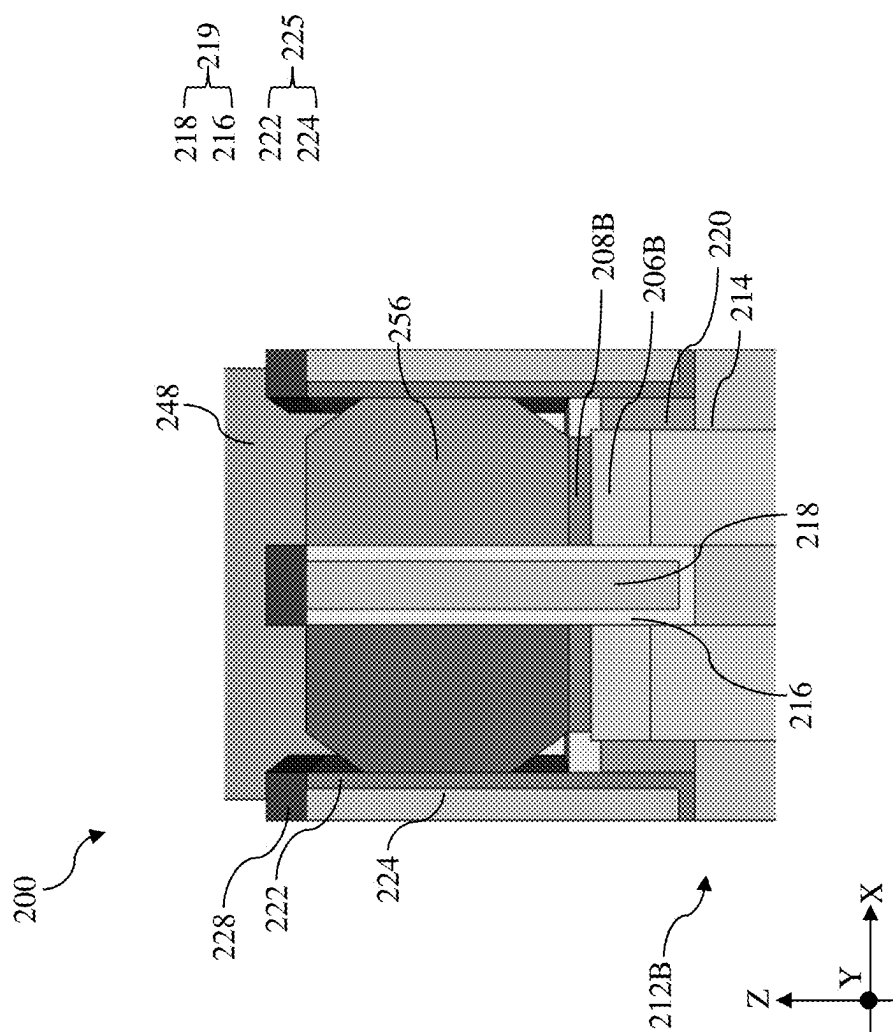


Fig. 14

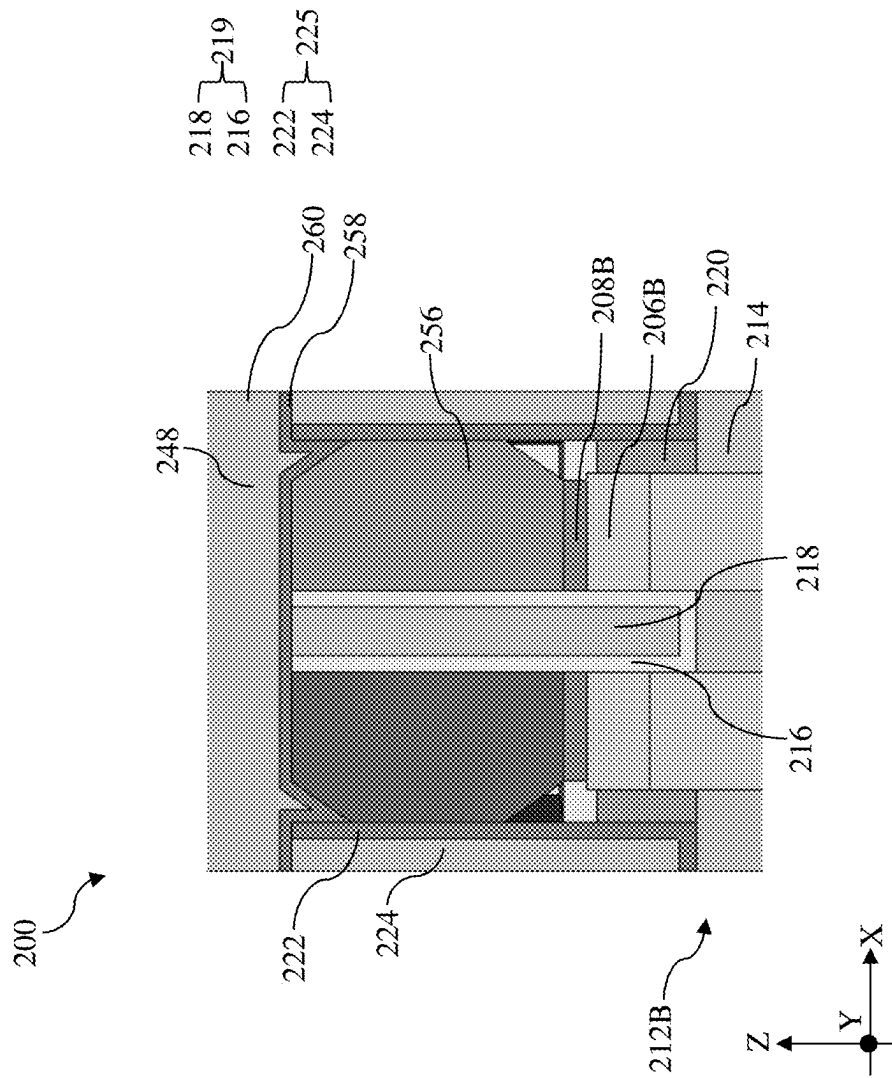


Fig. 15

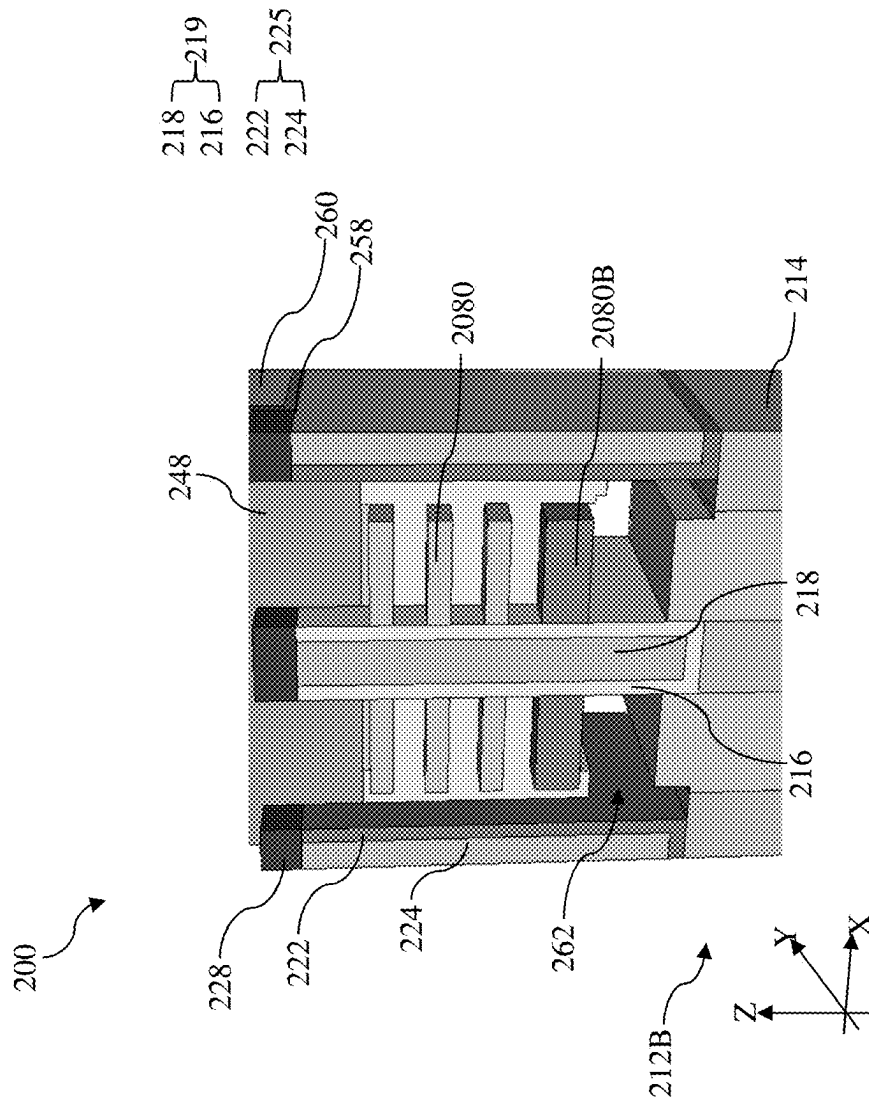


Fig. 16

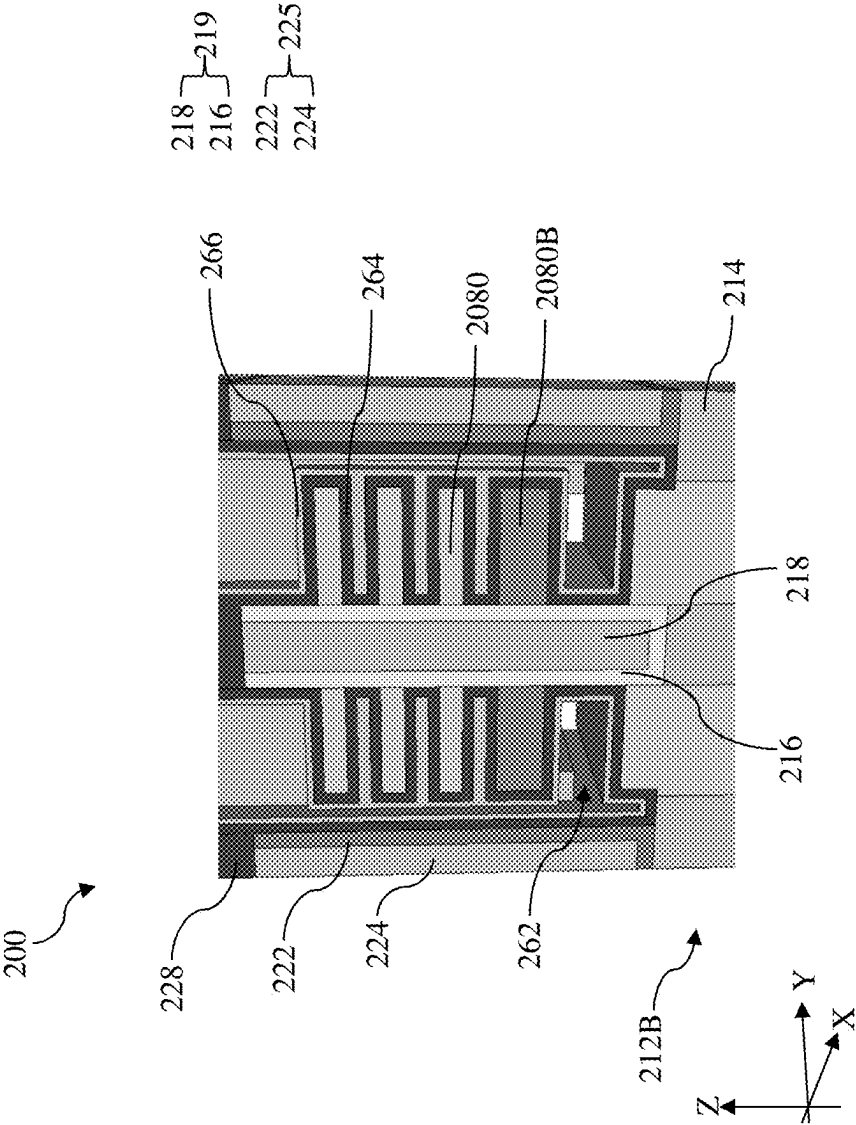


Fig. 17

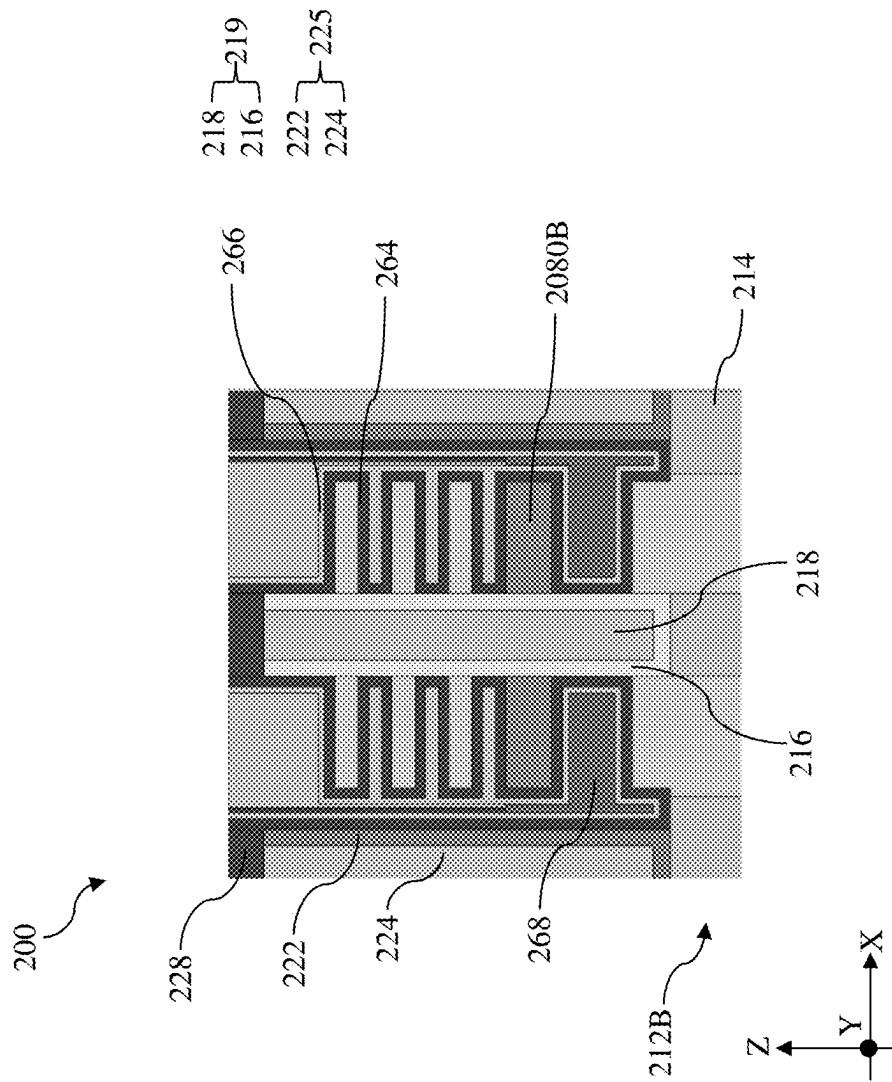


Fig. 18

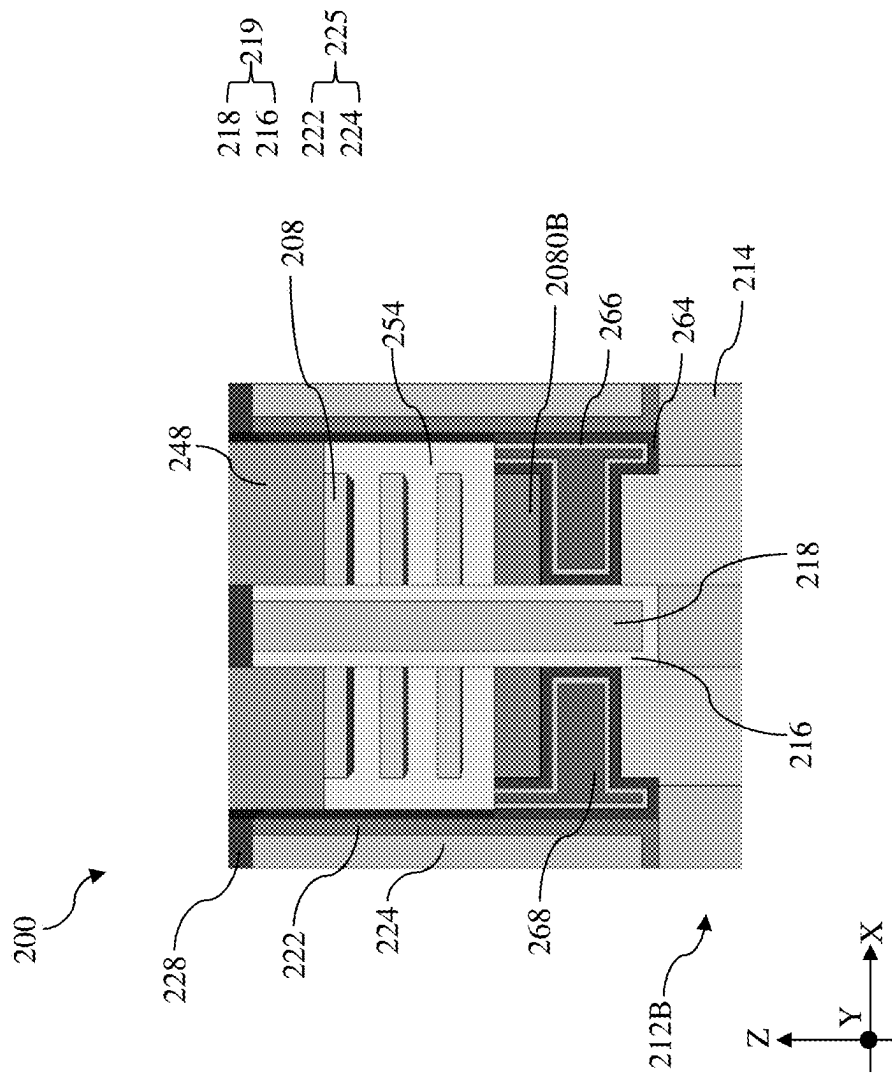


Fig. 19

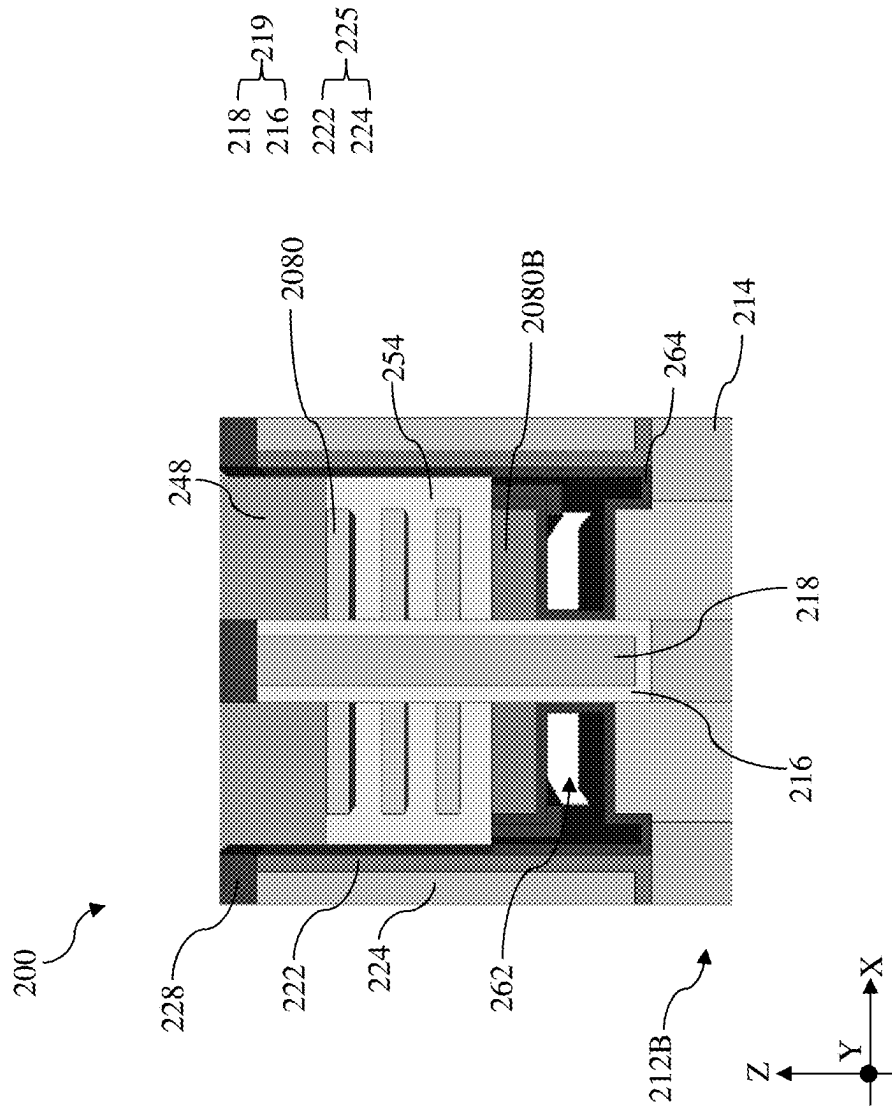


Fig. 20

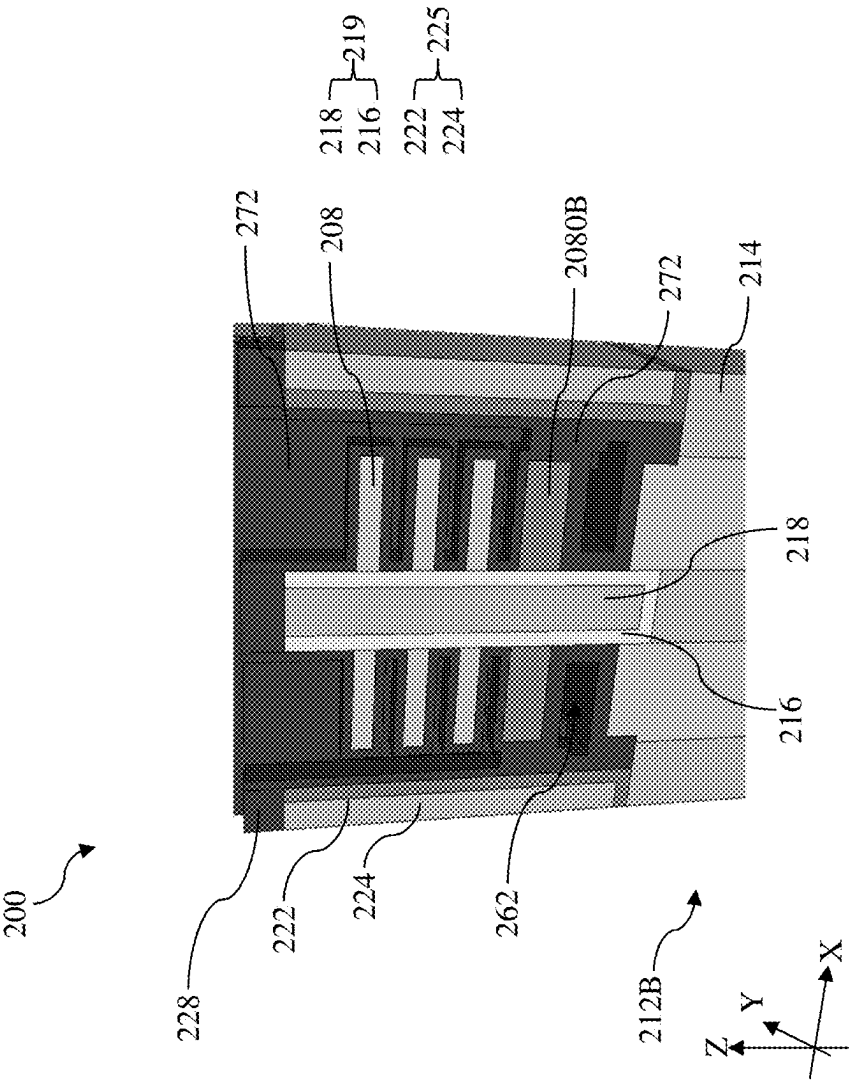
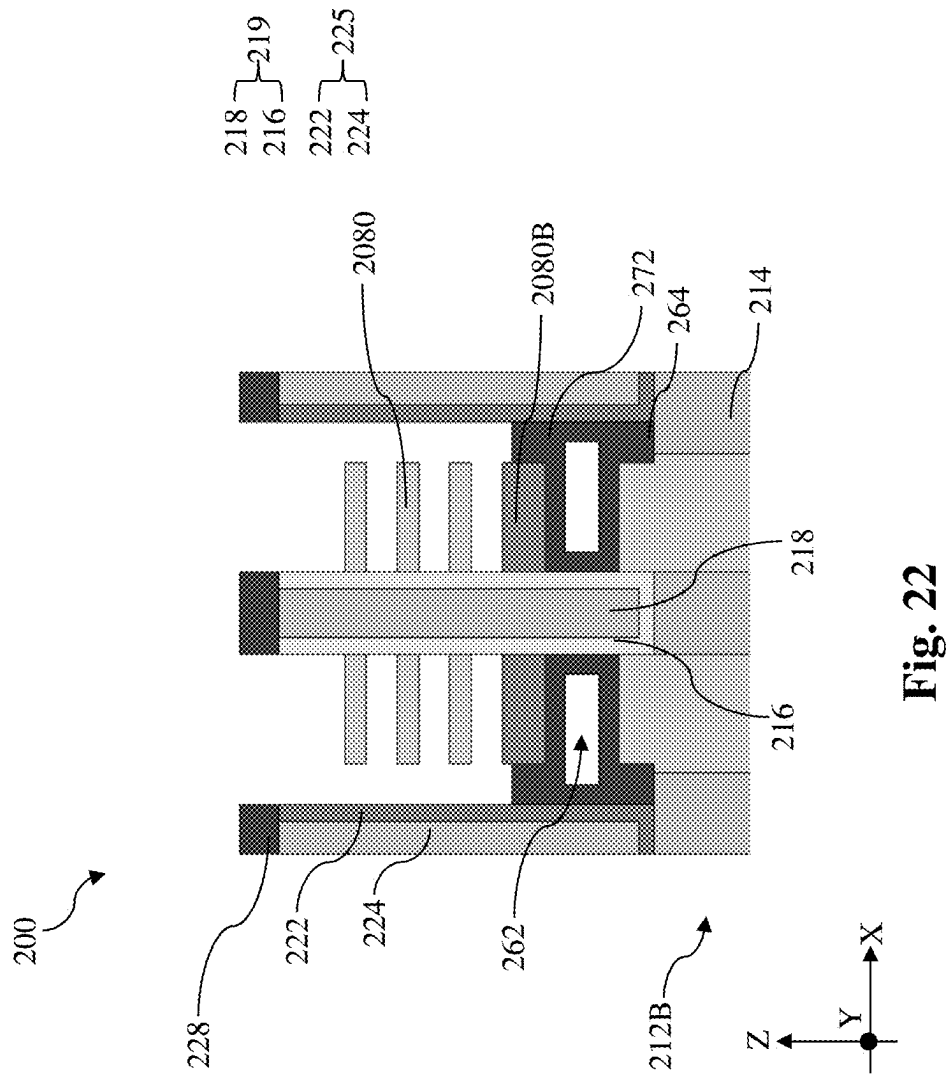


Fig. 21



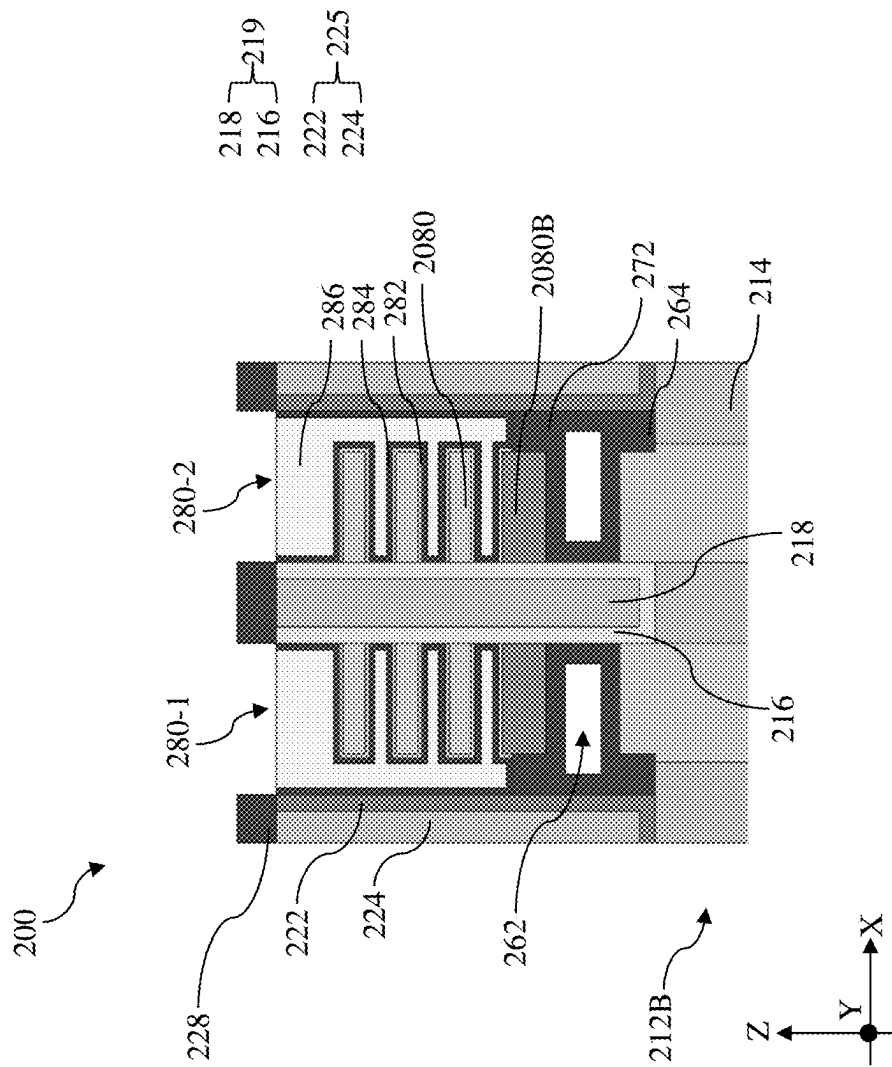
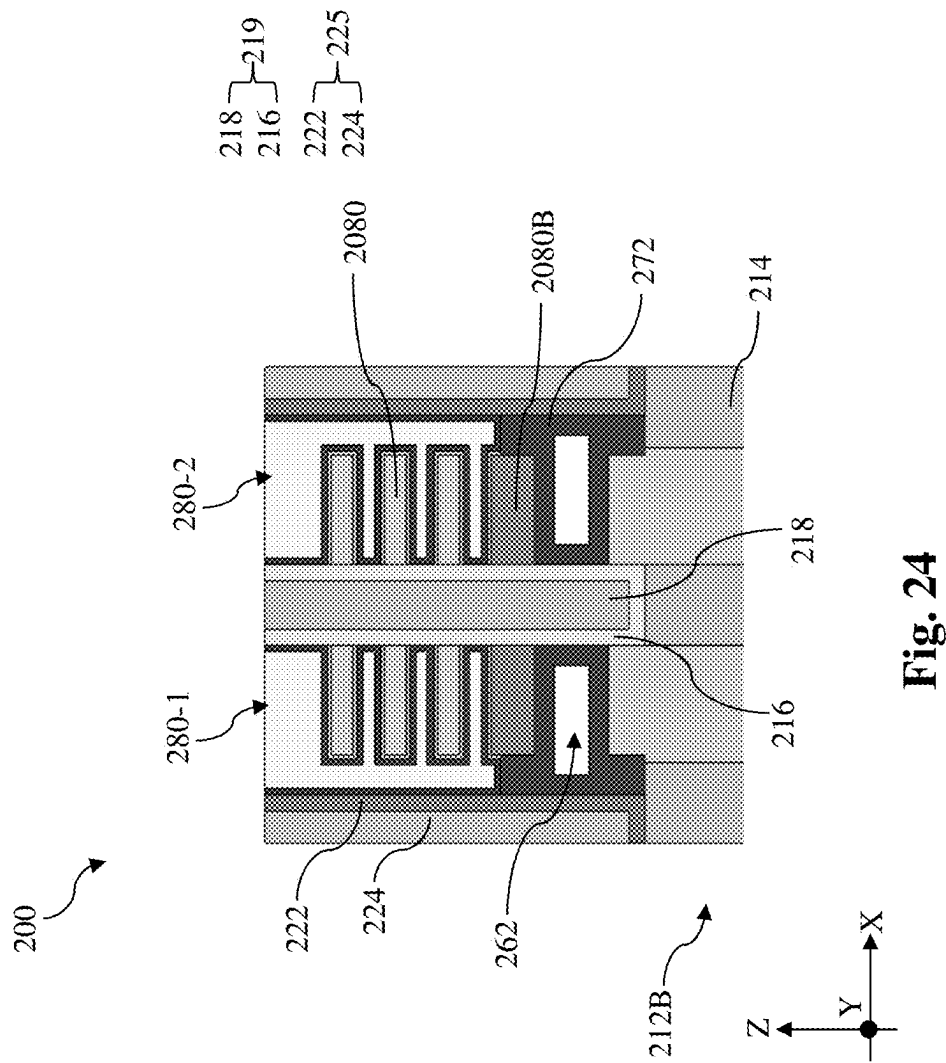


Fig. 23



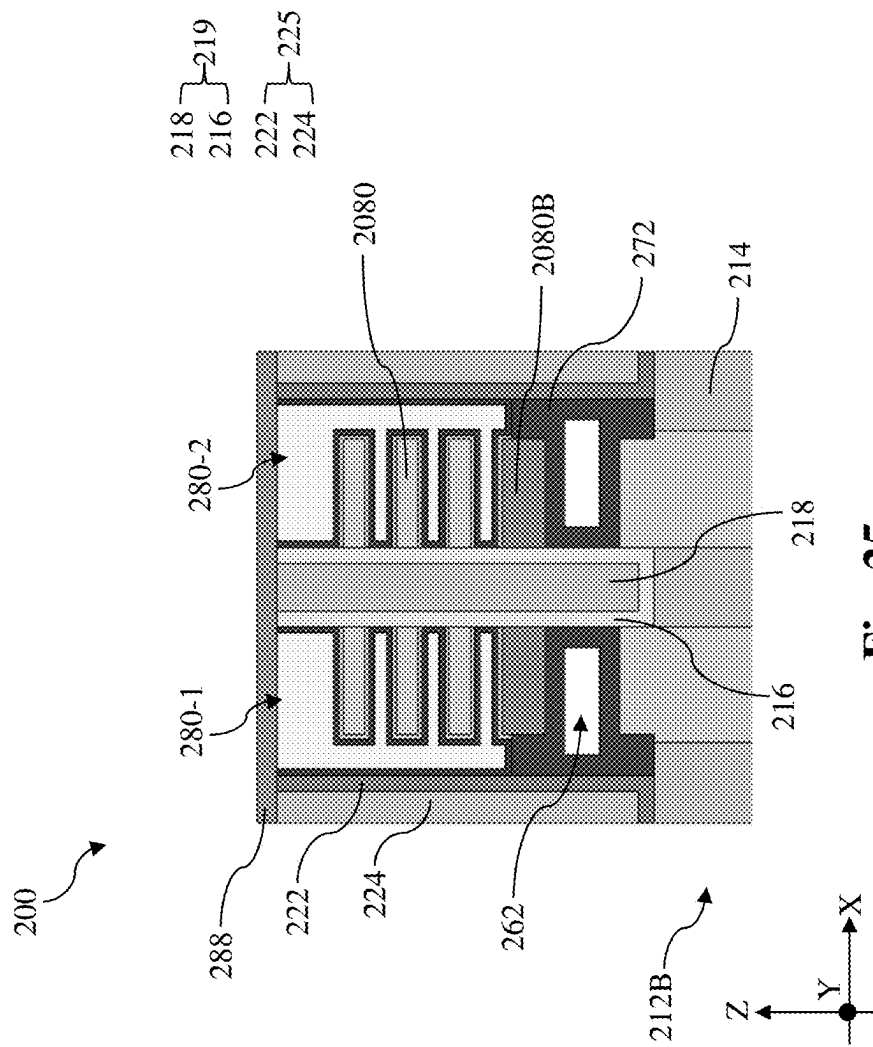


Fig. 25

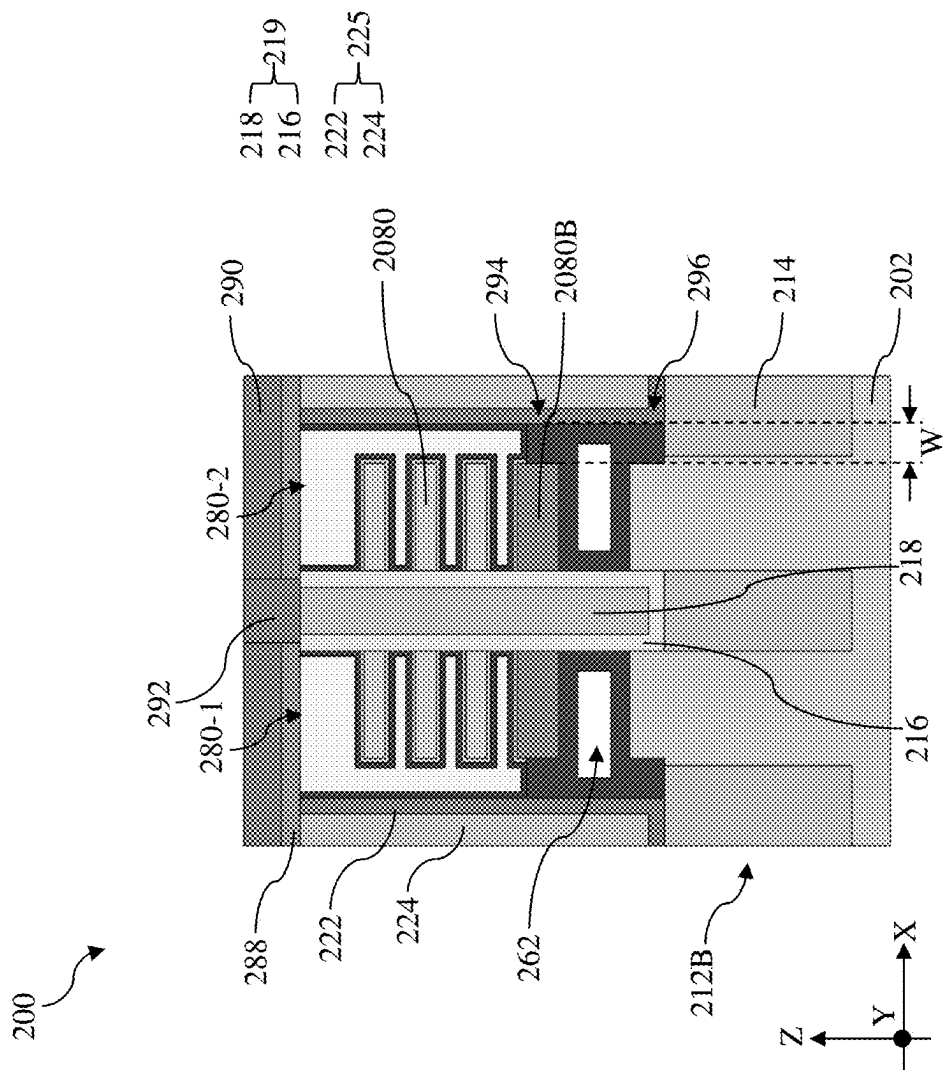


Fig. 26

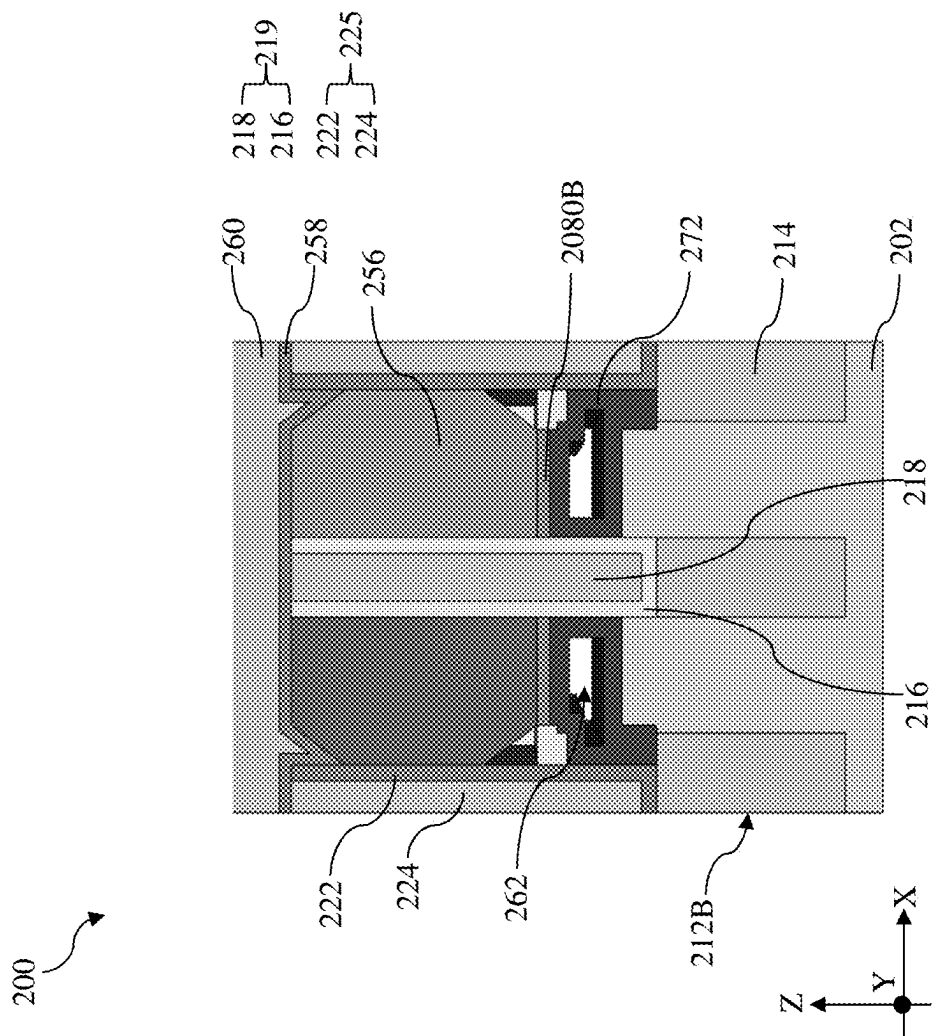


Fig. 27

1

GAP-INSULATED SEMICONDUCTOR DEVICE

PRIORITY DATA

This is a continuation of U.S. patent application Ser. No. 18/079,691, filed Dec. 12, 2022, which is continuation of U.S. patent application Ser. No. 17/142,640, filed Jan. 6, 2021, the entirety of which is hereby incorporated herein by reference.

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

For example, as integrated circuit (IC) technologies progress towards smaller technology nodes, multi-gate metal-oxide-semiconductor field effect transistor (multi-gate MOSFET, or multi-gate devices) have been introduced to improve gate control by increasing gate-channel coupling, reducing off-state current, and reducing short-channel effects (SCEs). A multi-gate device generally refers to a device having a gate structure, or portion thereof, disposed over more than one side of a channel region. Fin-like field effect transistors (FinFETs) and multi-bridge-channel (MBC) transistors are examples of multi-gate devices that have become popular and promising candidates for high performance and low leakage applications. A FinFET has an elevated channel wrapped by a gate on more than one side (for example, the gate wraps a top and sidewalls of a “fin” of semiconductor material extending from a substrate). An MBC transistor has a gate structure that can extend, partially or fully, around a channel region to provide access to the channel region on two or more sides. Because its gate structure surrounds the channel regions, an MBC transistor may also be referred to as a surrounding gate transistor (SGT) or a gate-all-around (GAA) transistor.

A channel of an MBC transistor includes a plurality of channel members that are formed from a fin-shaped structure rising from a substrate. Because the fin-shaped structure has a base portion that is connected with the substrate, the base portion may provide paths for leakage. While existing MBC transistor structures are generally adequate for their intended purposes, they are not satisfactory in all aspects.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

2

FIGS. 1A and 1B collectively illustrate a flowchart of a method for forming a semiconductor device, according to one or more aspects of the present disclosure.

FIGS. 2-27 illustrate fragmentary perspective views or cross-sectional views of a workpiece during various fabrication stages in the method of FIGS. 1A and 1B, according to one or more aspects of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range considering variations that inherently arise during manufacturing as understood by one of ordinary skill in the art. For example, the number or range of numbers encompasses a reasonable range including the number described, such as within $\pm 10\%$ of the number described, based on known manufacturing tolerances associated with manufacturing a feature having a characteristic associated with the number. For example, a material layer having a thickness of “about 5 nm” can encompass a dimension range from 4.25 nm to 5.75 nm where manufacturing tolerances associated with depositing the material layer are known to be $\pm 15\%$ by one of ordinary skill in the art. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

The present disclosure is generally related to isolation structures to reduce bulk leakage, and more particularly to a gap-containing structure to isolate active regions from the bulk substrate.

To improve drive current to meet design needs, MBC transistors may include nanoscale channel members that are thin and wide. Such MBC transistors may also be referred to as nanosheet transistors. While nanosheet transistors are able to provide satisfactory drive current and channel con-

trol, their wider nanosheet channel members may make it challenging to reduce cell sizes. In some example structures, fish-bone structures or fork-sheet structures may be implemented to reduce cell dimensions. In a fish-bone structure or a fork-sheet structure, adjacent stacks of channel members may be divided by dielectric fins (or hybrid fins). Regardless of the fish-bone or fork-sheet structures, close proximity between the channel members and the base portion may lead to leakage through the bulk substrate.

The present disclosure provides an isolation structure to vertically isolate the channel members from the bulk substrate to reduce bulk leakage. A semiconductor structure according to the present disclosure includes a first base portion and a second base portion arising from a substrate. An isolation feature is sandwiched between the first base portion and the second base portion. A center dielectric fin is disposed over the isolation feature. A first anti-punch-through (APT) feature over the first base portion and a second APT feature over the second base portion. A first stack of channel members is disposed over the first APT feature and a second stack of channel members is disposed over the second APT feature. The center dielectric fin is disposed between the first APT feature and the second APT feature as well as between the first stack of channel members and the second stack of channel members. A first cavity is disposed between the first base portion and the first APT feature. A second cavity is disposed between the second base portion and the second APT feature. The first cavity, the first APT feature, the second cavity, and the second APT feature isolate the channel members from the base portions to reduce or eliminate bulk leakage.

The various aspects of the present disclosure will now be described in more detail with reference to the figures. FIGS. 1A and 1B collectively illustrate a flowchart of a method 100 of forming a semiconductor device. Method 100 is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated in method 100. Additional steps may be provided before, during and after method 100, and some steps described can be replaced, eliminated, or moved around for additional embodiments of the methods. Not all steps are described herein in detail for reasons of simplicity. Method 100 is described below in conjunction with FIGS. 2-27, which illustrate fragmentary cross-sectional views of a workpiece 200 at different stages of fabrication according to embodiments of method 100. Because a semiconductor device will be formed from the workpiece 200, the workpiece 200 may be referred to as a semiconductor device 200 as the context requires. Although embodiments that include fish-bone or fork-sheet transistors are illustrated in the figures, the present disclosure is not so limited and may be applicable to other multi-gate devices, such as MBC transistors or FinFETs. Throughout FIGS. 2-27, the X direction, the Y direction, and the Z direction are perpendicular to one another and are used consistently. Additionally, throughout the present disclosure, like reference numerals are used to denote like features.

Referring to FIGS. 1A and 2, method 100 includes a block 102 where a workpiece 200 is received. As shown in FIG. 2, the workpiece 200 includes a substrate 202 and a stack 204 disposed on the substrate 202. In one embodiment, the substrate 202 may be a silicon (Si) substrate. In some other embodiments, the substrate 202 may include other semiconductor materials such as germanium (Ge), silicon germanium (SiGe), or a III-V semiconductor material. Example III-V semiconductor materials may include gallium arsenide (GaAs), indium phosphide (InP), gallium phosphide (GaP), gallium nitride (GaN), gallium arsenide phosphide (GaAsP),

aluminum indium arsenide (AlInAs), aluminum gallium arsenide (AlGaAs), gallium indium phosphide (GaInP), and indium gallium arsenide (InGaAs). The substrate 202 may include multiple n-type well regions and multiple p-type well regions. A p-type well region may be doped with a p-type dopant (i.e., boron (B)). An n-type well region may be doped with an n-type dopant (i.e., phosphorus (P) or arsenic (As)).

In some embodiments represented in FIG. 2, the stack 204 may include a bottom sacrificial layer 206B over the substrate 202, an anti-punch-through (APT) layer 208B over the bottom sacrificial layer 206B, alternating channel layers 208 and sacrificial layers 206 over the bottom sacrificial layer 206B, and a top sacrificial layer 206T over the sacrificial layers 206 and the channel layers 208. The bottom sacrificial layer 206B, the APT layer 208B, the top sacrificial layer 206T, the sacrificial layers 206 and the channel layers 208 may be deposited using an epitaxial process. Example epitaxial process may include vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), molecular beam epitaxy (MBE), and/or other suitable processes. The channel layers 208 and the sacrificial layers 206 may have different semiconductor compositions. In some implementations, the channel layers 208 are formed of silicon (Si) and sacrificial layers 206 are formed of silicon germanium (SiGe). The additional germanium (Ge) content in the sacrificial layers 206 allow selective removal or recess of the sacrificial layers 206 without substantial damages to the channel layers 208. The sacrificial layers 206 and the channel layers 208 are disposed alternately such that sacrificial layers 206 interleave the channel layers 208. FIG. 2 illustrates that three (3) layers of the sacrificial layers 206 and three (3) layers of the channel layers 208 are alternately and vertically arranged, which is for illustrative purposes only and not intended to be limiting beyond what is specifically recited in the claims. The number of layers depends on the desired number of channels members 208 for the semiconductor device 200. In some embodiments, the number of the channel layers 208 is between 1 and 6. The bottom sacrificial layer 206B may be formed of silicon germanium (SiGe). Different from the sacrificial layers 206, a germanium content of the bottom sacrificial layer 206B may be smaller than a germanium content of the sacrificial layers 206. In some embodiments, the germanium content of the sacrificial layers 206 may be between about 20% and about 30% and the germanium content of the bottom sacrificial layer 206B may be about 80% to about 90% of the germanium content of the sacrificial layer 206. In some instances, the germanium content of the bottom sacrificial layer 206B may be between about 16% and about 27%. The smaller germanium content of the bottom sacrificial layer 206B allows the sacrificial layers 206 to be selectively removed without substantially damaging the bottom sacrificial layer 206B. The bottom sacrificial layer 206B is thicker than each of the sacrificial layers 206 along the Z direction. As will be described below, removal of the bottom sacrificial layer 206B allows formation of a cavity that is large enough not to be filled up by a first dielectric layer 264 and a liner 266. In some instances, each of the sacrificial layers 206 may have a thickness between about 4 nm and about 15 nm while the bottom sacrificial layer 206B may have a thickness between about 8 and about 30 nm.

The APT layer 208B may include silicon (Si) and may be doped with a dopant having a conductivity type different from the conductivity type of the desired MBC transistor. For example, when a p-type MBC transistor is desired, the APT layer 208B may be doped with an n-type dopant, such

5

as phosphorus (P) or arsenic (As). When an n-type MBC transistor is desired, the APT layer **208B** may be doped with a p-type dopant, such as boron (B) or boron difluoride (BF₂). The deposition of the APT layer **208B** may be followed by an activation step, which may include annealing. After activation, the activation concentration of the dopants in the APT layer **208B** may be greater than 1×10^{18} atoms/cm³, such as between about 1×10^{18} atoms/cm³ and about 5×10^{20} atoms/cm³. The APT layer **208B** is not intended to be formed into a channel member. In that regard, the counter-doping (having a dopant conductivity type different from the dopant conductivity type in the source/drain feature) in the APT layer **208B** functions to disable the APT member **2080B** (to be described below) formed from the APT layer **208B**. The activation concentration in the APT layer **208B** may be detected by energy dispersive X-Ray spectroscopy (EDS). The APT layer **208B** may be thicker than each of the channel layers **208** along the Z direction. As will be described below, the greater thickness of the APT layer **208B** allows it to withstand etching during recess of source/drain regions of the fin-shaped structures **212**. In some extreme cases, the APT layer **208B** and the channel layers **208** may have the same thickness if subsequent etching processes are more selective and cause less collateral damages to the APT layer **208B**. In some instances, each of the channel layers **208** may have a thickness between about 8 nm and about 20 nm while the APT layer **208B** may have a thickness between about 8 and about 30 nm.

Like the sacrificial layers **206**, the top sacrificial layer **206T** may be formed of silicon germanium (SiGe). In some instances, compositions of the sacrificial layers **206** and the top sacrificial layer **206T** are substantially the same. The top sacrificial layer **206T** may be thicker than the other sacrificial layers **206** and functions to protect the stack **204** from damages during fabrication processes. In some instances, a thickness of the top sacrificial layer **206T** may be between about 20 nm and about 40 nm while a thickness of a sacrificial layer **206** may be between about 4 nm and about 15 nm.

Referring to FIGS. 1A and 3, method **100** includes a block **104** where the stack **204** and the substrate **202** are patterned to form fin-shaped structures **212** separated by a center trench **211C** and separation trenches **211**. To pattern the stack **204** and the substrate **202**, a fin-top hard mask layer **210** is deposited over the top sacrificial layer **206T**. The fin-top hard mask layer **210** is then patterned to serve as an etch mask to pattern the stack **204** and a portion of the substrate **202**. In some embodiments, the fin-top hard mask layer **210** may be deposited using CVD, plasma-enhanced CVD (PECVD), atomic layer deposition (ALD), plasma-enhanced ALD (PEALD), or a suitable deposition method. The fin-top hard mask layer **210** may be a single layer or a multilayer. When the fin-top hard mask layer **210** is a multi-layer, the fin-top hard mask layer **210** may include a pad oxide and a pad nitride layer. In an alternative embodiment, the fin-top hard mask layer **210** is a single layer and is formed of silicon (Si). The fin-shaped structures **212** may be patterned using suitable processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a material layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned material layer using a self-aligned process. The

6

material layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fin-top hard mask layer **210** and then the patterned fin-top hard mask layer **210** may be used as an etch mask to etch the stack **204** and the substrate **202** to form fin-shaped structures **212**. The etching process can include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes.

As shown in FIG. 3, each of the fin-shaped structures **212** includes a base portion **212B** formed from a portion of the substrate **202** and a top portion **212T** formed from the stack **204**. The top portion **212T** is disposed over the base portion **212B**. The fin-shaped structures **212** extend lengthwise along the Y direction and extend vertically along the Z direction from the substrate **202**. Along the X direction, the two fin-shaped structures **212** in FIG. 3 are separated from one another by the center trench **211C** while they are separated from other adjacent fin-shaped structures by separation trenches **211**. A width of the separation trenches **211** is greater than a width of the center trench **211C** along the X direction. In some embodiments, a width of the center trench **211C** is between about 10 nm and about 20 nm and a width of the separation trench **211** is between about 20 nm and about 40 nm. In some implementations, the separation trenches **211** are disposed over a junction of an n-type well region and a p-type well region and may therefore be referred to as junction trenches **211**. In those implementations, the greater width of the separation trenches **211** functions to separate devices of different conductivity types.

Referring to FIGS. 1A and 4, method **100** includes a block **106** where an isolation feature **214** is formed in the center trench **211C** and the separation trenches **211**. The isolation feature **214** may be referred to as a shallow trench isolation (STI) feature **214**. In an example process to form these isolation feature **214**, a dielectric material is deposited over the workpiece **200**, filling the center trench **211C** and the separation trenches **211** with the dielectric material. In some embodiments, the dielectric material may tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. In various examples, at block **106**, the dielectric material may be deposited by flowable CVD (FCVD), spin-on coating, and/or other suitable process. The deposited dielectric material is then thinned and planarized, for example by a chemical mechanical polishing (CMP) process, until the top sacrificial layer **206T** is exposed. After the planarization, the deposited dielectric material is etched back until the top portions **212T** of the fin-shaped structures **212** rises above the isolation feature **214**. In some embodiments, a portion of the base portion **212B** may also rise above the isolation feature **214**. At this point, the base portions **212B**, or a substantial portion thereof, is surrounded by the isolation features **214**. The isolation feature **214** reduces the depths of the center trench **211C** and the separation trenches **211**.

Referring to FIGS. 1A and 5, method **100** includes a block **108** where a center dielectric fin **219** is formed. To form the center dielectric fin **219**, a first layer **216** and a second layer **218** are conformally deposited over the workpiece **200**, including in the center trench **211C** and the separation trenches **211**. The first layer **216** may be conformally deposited using CVD, ALD, or a suitable method. The first layer **216** lines the sidewalls and the bottom surfaces of the center trench **211C** and the separation trenches **211**. The second layer **218** is then conformally deposited over the first layer **216** using CVD, high density plasma CVD (HDPCVD),

7

and/or other suitable process. In some instances, a dielectric constant of the second layer **218** is smaller than that of the first layer **216**. The first layer **216** may include silicon nitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, aluminum nitride, aluminum oxynitride, zirconium nitride, silicon oxynitride, or a suitable dielectric material. In one embodiment, the first layer **216** includes silicon oxycarbonitride. The second layer **218** may include silicon oxide or a suitable dielectric material. In one embodiment, the second layer **218** includes silicon oxide. The conformally deposited first layer **216** and second layer **218** are etched back to expose the top sacrificial layer **206T**. Due to the loading effect, the deposited first layer **216** and the second layer **218** in the wider separation trenches **211** are removed by the etch back process while the deposited first layer **216** and the second layer **218** in the narrower center trench **211C** remains to become the center dielectric fin **219**. In some embodiments, the first layer **216** and the second layer **218** may be etched back in a dry etch process that uses oxygen, nitrogen, a fluorine-containing gas (e.g., CF_4 , SF_6 , CH_2F_2 , CHF_3 , and/or C_2F_6), a chlorine-containing gas (e.g., Cl_2 , CHCl_3 , CCl_4 , and/or BCl_3), a bromine-containing gas (e.g., HBr and/or CHBr_3), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof. In some implementations, the etch back may include a first stage that is directed toward the second layer **218** and a second stage that is directed toward the first layer **216**. As shown in FIG. 5, upon conclusion of the etch back, the isolation feature **214** is exposed in the separation trenches **211**.

Referring to FIGS. 1A and 6, method **100** includes a block **110** where a cladding layer **220** is formed over the fin-shaped structures **212**. With the separation trenches **211** exposed, a cladding layer **220** is deposited over the workpiece **200**, including over the sidewalls of the separation trenches **211**. In some embodiments, the cladding layer **220** may have a composition similar to that of the sacrificial layers **206** or the top sacrificial layer **206T**. In one example, the cladding layer **220** may be formed of silicon germanium (SiGe). Their common composition allows selective and simultaneous removal of the sacrificial layers **206** and the cladding layer **220** in a subsequent process. In some embodiments, the cladding layer **220** may be conformally and epitaxially grown using vapor phase epitaxy (VPE) or molecular beam epitaxy (MBE). As shown in FIG. 6, the cladding layer **220** is selectively disposed on exposed sidewall surfaces in the separation trenches **211**, but not on the isolation feature **214** or the center dielectric fin **219**. After the deposition of the cladding layer **220**, an etch back process is performed to remove the cladding layer **220** over the top sacrificial layer **206T** and the center dielectric fin **219**.

Referring to FIGS. 1A and 7, method **100** includes a block **112** where separation dielectric fins **225** are formed. To form the separation dielectric fins **225**, a third layer **222** and a fourth layer **224** are conformally deposited into the separation trenches **211**. The composition and the formation of the third layer **222** may be similar to those of the first layer **216**. The fourth layer **224** may include tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. In some instances, the fourth layer **224** may include silicon oxide. In some embodiments, the fourth layer **224** may be deposited by flowable CVD (FCVD), spin-on coating, and/or other suitable process. In one embodiment, the fourth layer **224** may be deposited using FCVD. After the

8

deposition of the third layer **222** and the fourth layer **224**, the workpiece **200** is planarized using a chemical mechanical polishing (CMP) process to expose the top sacrificial layer **206T**. The planarized third layer **222** and the fourth layer **224** in the separation trenches **211** may be collectively referred to as separation dielectric fins **225**.

Referring to FIGS. 1A and 8, method **100** includes a block **114** where a helmet layer **228** is formed over each of the center dielectric fin **219** and the separation dielectric fins **225**. At block **114**, the center dielectric fin **219** and the separation dielectric fins **225** are selectively etched back to form recesses and a helmet layer **228** is deposited in such recesses. In some embodiments, the selective etch back may be performed using a dry etch process that may include oxygen (O_2), nitrogen (N_2), a fluorine-containing gas (e.g., CF_4 , SF_6 , CH_2F_2 , CHF_3 , and/or C_2F_6), a chlorine-containing gas (e.g., Cl_2 , CHCl_3 , CCl_4 , and/or BCl_3), a bromine-containing gas (e.g., HBr and/or CHBr_3), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof. The helmet layer **228** may include aluminum oxide, zirconium oxide, zirconium aluminum oxide, hafnium oxide, other metal oxide, or a combination thereof. In one embodiment, the helmet layer **228** may include hafnium oxide. In some embodiments, the helmet layer **228** may be deposited using CVD, HDPCVD, or a suitable deposition technique. After the deposition of the helmet layer **228**, the workpiece **200** is planarized using a CMP process to remove excess helmet layer **228** on the cladding layer **220** and the top sacrificial layer **206T**.

Referring to FIGS. 1A and 9, method **100** includes a block **116** where the top sacrificial layer **206T** is removed from the fin-shaped structures **212**. At block **116**, the workpiece **200** is etched to selectively remove a portion of the cladding layer **220** and the top sacrificial layer **206T** to expose the topmost channel layer **208**, without substantially damaging the helmet layer **228**. Because the top sacrificial layer **206T** and the cladding layer **220** are formed of silicon germanium (SiGe), the etch process at block **116** may be selective to silicon germanium (SiGe). In some instances, the cladding layer **220** and the top sacrificial layer **206T** may be etched using a selective wet etch process that includes ammonium hydroxide (NH_4OH), hydrogen fluoride (HF), hydrogen peroxide (H_2O_2), or a combination thereof. As shown in FIG. 9, after the removal of the top sacrificial layer **206T** and the etching of the cladding layer **220**, the center dielectric fin **219** and the separation dielectric fins **225** rise above the topmost channel layer **208**.

Referring to FIGS. 1A, 10 and 11, method **100** includes a block **118** where a dummy gate stack **240** is formed over the channel regions of the fin-shaped structures **212**. In some embodiments, a gate replacement process (or gate-last process) is adopted where the dummy gate stack **240** serves as a placeholder for a functional gate structure. Other processes and configuration are possible. As shown in FIG. 10, the dummy gate stack **240** includes a dummy dielectric layer **236** and a dummy electrode **238** disposed over the dummy dielectric layer **236**. For patterning purposes, a gate top hard mask **246** is deposited over the dummy gate stack **240**. The gate top hard mask **246** may be a multi-layer and include a silicon nitride mask layer **242** and a silicon oxide mask layer **244** over the silicon nitride mask layer **242**. The regions of the fin-shaped structures **212** underlying the dummy gate stack **240** may be referred to as channel regions. Each of the channel regions in a fin-shaped structure **212** is sandwiched between two source/drain regions for source/drain formation. In an example process, the dummy dielectric layer **236** is blanketly deposited over the workpiece **200** by CVD. A

semiconductor layer for the dummy electrode **238** is then blanketly deposited over the dummy dielectric layer **236**. The dummy dielectric layer **236** and the semiconductor layer for the dummy electrode **238** are then patterned using photolithography processes to form the dummy gate stack **240**. In some embodiments, the dummy dielectric layer **236** may include silicon oxide and the dummy electrode **238** may include polycrystalline silicon (polysilicon).

Reference is made to FIG. **11**. At block **118**, at least one gate spacer **248** is formed along sidewalls of the dummy gate stacks **240**. Dielectric materials for the at least one gate spacer **248** may be selected to allow selective removal of the dummy gate stack **240**. Suitable dielectric materials may include silicon nitride, silicon oxycarbonitride, silicon carbonitride, silicon oxide, silicon oxycarbide, silicon carbide, silicon oxynitride, and/or combinations thereof. In an example process, the at least one gate spacer **248** may be conformally deposited over the workpiece **200** using CVD, subatmospheric CVD (SACVD), or ALD.

Referring to FIGS. **1A** and **11**, method **100** includes a block **120** where the source/drain regions of the fin-shaped structures **212** are recessed to form source/drain recesses **250**. With the dummy gate stack **240** and the at least one gate spacer **248** serving as an etch mask, the workpiece **200** is anisotropically etched to form the source/drain recesses **250** (or source/drain trenches **250**) over the source/drain regions of the fin-shaped structures **212**. In some embodiments as illustrated in FIG. **11**, operations at block **120** may completely remove the sacrificial layers **206** and channel layers **208** in the source/drain regions. In the depicted embodiments, the recess at block **120** may also remove a top portion of the APT layer **208B**, thereby thinning it. The anisotropic etch at block **120** may include a dry etch process. For example, the dry etch process may implement hydrogen (H_2), a fluorine-containing gas (e.g., CF_4 , SF_6 , CH_2F_2 , CHF_3 , and/or C_2F_6), a chlorine-containing gas (e.g., Cl_2 , $CHCl_3$, CCl_4 , and/or BCl_3), a bromine-containing gas (e.g., HBr and/or $CHBr_3$), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof. In FIG. **11**, a lower portion of the APT layer **208B** and the bottom sacrificial layer **206B** remains unetched and may remain sandwiched between the cladding layer **220** and the center dielectric fin **219**. Sidewalls of the channel layers **208**, the sacrificial layers **206**, and the cladding layer **220** are exposed in the source/drain recesses **250**.

Referring to FIGS. **1A**, **12** and **13**, method **100** includes a block **122** where inner spacer features **254** are formed. Referring to FIG. **12**, at block **122**, the sacrificial layers **206** and the cladding layer **220** exposed in the source/drain trenches **250** are first selectively and partially recessed to form inner spacer recesses **252**, while the exposed channel layers **208** and the bottom sacrificial layer **206B** are substantially unetched. In an embodiment where the channel layers **208** consist essentially of silicon (Si) and sacrificial layers **206** and the cladding layer **220** consist essentially of silicon germanium (SiGe), the selective and partial recess of the sacrificial layers **206** and the cladding layer **220** may include a SiGe oxidation process followed by a SiGe oxide removal. In that embodiments, the SiGe oxidation process may include use of ozone. In some other embodiments, the selective recess may include a selective isotropic etching process (e.g., a selective dry etching process or a selective wet etching process), and the extent at which the sacrificial layers **206** and the cladding layer **220** are recessed is controlled by duration of the etching process. The selective dry etching process may include use of one or more fluorine-based etchants, such as fluorine gas or hydrofluorocarbons.

The selective wet etching process may include ammonium hydroxide (NH_4OH), hydrogen fluoride (HF), hydrogen peroxide (H_2O_2), or a combination thereof (e.g. an APM etch that includes an ammonia hydroxide-hydrogen peroxide-water mixture). As described above, the bottom sacrificial layer **206B** includes a lower germanium content than the sacrificial layers **206** (as well as that of the cladding layer **220**) and that allows the selective recess of the sacrificial layers **206** and the cladding layer **220**. After the formation of the inner spacer recesses **252**, an inner spacer material layer is then conformally deposited using CVD or ALD over the workpiece **200**, including over and into the inner spacer recesses **252** and the space left behind by the removed portion of the cladding layer **220**. The inner spacer material may include silicon nitride, silicon oxycarbonitride, silicon carbonitride, silicon oxide, silicon oxycarbide, silicon carbide, or silicon oxynitride. After the deposition of the inner spacer material layer, the inner spacer material layer is etched back to form inner spacer features **254**, as illustrated in FIG. **13**.

Referring to FIGS. **1A** and **14**, method **100** includes a block **124** where source/drain features **256** are formed. The source/drain features **256** are selectively and epitaxially deposited on the exposed semiconductor surfaces of the channel layers **208**, the APT layer **208B**, and the substrate **202** in the source/drain trenches **250**. The source/drain features **256** may be deposited using an epitaxial process, such as vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), molecular beam epitaxy (MBE), and/or other suitable processes. Depending on the design of the semiconductor device **200**, the source/drain features **256** may be n-type or p-type. When the source/drain features **256** are n-type, they may include silicon (Si) doped with an n-type dopant, such as phosphorus (P) or arsenic (As). When the source/drain features **256** are p-type, they may include silicon germanium (SiGe) doped with a p-type dopant, such as boron (B) or gallium (Ga). Doping of the source/drain features **256** may be performed either in situ with their deposition or ex situ using an implantation process, such as a junction implant process.

Referring still to FIGS. **1A** and **15**, method **100** includes a block **126** where a contact etch stop layer (CESL) **258** and an interlayer dielectric (ILD) layer **260** are deposited. In some embodiments, the helmet layer **228** in the source/drain regions are selectively removed before the deposition of the CESL **258**. In some instances, the helmet layer **228** may be selectively etched away using buffered hydrofluoric acid (BHF) or diluted hydrofluoric acid (DHF). After the removal of the helmet **228**, the CESL **258** is first conformally deposited over the workpiece **200** and then the ILD layer **260** is blanketly deposited over the CESL **258**. The CESL **258** may include silicon nitride, silicon oxide, silicon oxynitride, and/or other materials known in the art. The CESL **258** may be deposited using ALD, plasma-enhanced chemical vapor deposition (PECVD) process and/or other suitable deposition or oxidation processes. In some embodiments, the ILD layer **260** includes materials such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. The ILD layer **260** may be deposited by spin-on coating, an FCVD process, or other suitable deposition technique. In some embodiments, after formation of the ILD layer **260**, the workpiece **200** may be annealed to improve integrity of the ILD layer **260**. To remove excess materials and to expose top surfaces of the dummy electrode **238** of

11

the dummy gate stacks **240**, a planarization process (such as a chemical mechanical polishing (CMP) process) may be performed to the workpiece **200** to provide a planar top surface. Top surfaces of the dummy electrodes **238** are exposed on the planar top surface.

Referring to FIGS. **1A** and **16**, method **100** includes a block **128** where the channel layers **208** and the APT layer **208B** in the channel regions are released as channel members **2080** and an APT member **2080B**. Referring to FIG. **16**, the dummy gate stack **240** exposed at the conclusion of block **126** is removed from the workpiece **200** by a selective etch process. The selective etch process may be a selective wet etch process, a selective dry etch process, or a combination thereof. In the depicted embodiments, the selective etch process selectively removes the dummy dielectric layer **236** and the dummy electrode **238** without substantially damaging the helmet layer **228** and the at least one gate spacer **248**. After the removal of the dummy gate stack **240**, channel layers **208**, sacrificial layers **206**, the APT layer **208B**, the bottom sacrificial layer **206B**, and the cladding layer **220** in the channel region are exposed. The exposed sacrificial layers **206**, the bottom sacrificial layer **206B**, and the cladding layer **220** may be selectively removed to release the channel layers **208** as channel members **2080** and release the APT layer **208B** as the APT member **2080B**. As shown in FIG. **16**, when viewed along the Y direction, the channel members **2080** and the APT member **2080B** have appearances of cantilever beams stemming from the center dielectric fin **219**. In the depicted embodiments where the channel members **2080** resemble a sheet or a nanosheet, the channel member release process may also be referred to as a sheet formation process. After their release, the channel members **2080** and the APT member **2080B** are in contact with the center dielectric fin **219** but are spaced apart from the separation dielectric fins **225**. The channel members **2080** and the APT member **2080B** are vertically stacked along the Z direction. The selective removal of the sacrificial layers **206**, the bottom sacrificial layer **206B** and the cladding layer **220** may be implemented by selective dry etch, selective wet etch, or other selective etch processes. In some embodiments, the selective wet etching includes ammonium hydroxide (NH_4OH), hydrogen fluoride (HF), hydrogen peroxide (H_2O_2), or a combination thereof (e.g. an APM etch that includes an ammonia hydroxide-hydrogen peroxide-water mixture). In some alternative embodiments, the selective removal includes silicon germanium oxidation followed by a silicon germanium oxide removal. For example, the oxidation may be provided by ozone clean and then silicon germanium oxide removed by an etchant such as NH_4OH . In the depicted embodiment, the removal of the bottom sacrificial layer **206B** and the cladding layer **220** also forms a cavity **262** below the released APT member **2080B**.

Referring to FIGS. **1A** and **17**, method **100** includes a block **130** where a first dielectric layer **264** is deposited around the channel members **2080** and the APT member **2080B**. In some embodiments, the first dielectric layer **264** is formed of a high-dielectric-constant (i.e., high-k) dielectric material with a dielectric constant greater than that of silicon dioxide, which is about 3.9. In some instances, the first dielectric layer **264** may include aluminum oxide, zirconium oxide, zirconium aluminum oxide, hafnium oxide, other metal oxide, or a combination thereof. In one embodiment, the first dielectric layer **264** may be formed of hafnium oxide. The first dielectric layer **264** may be deposited using ALD. In some implementations, the first dielectric layer **264** may have a thickness between about 1.5 nm and about 4 nm.

12

Referring to FIGS. **1B** and **17**, method **100** includes a block **132** where a liner **266** is deposited over the first dielectric layer **264**. In some embodiments, the liner **266** is formed of silicon oxide. The liner **266** may be deposited over the first dielectric layer **264** by ALD to a thickness between about 1.5 nm and about 4 nm to plug the space between adjacent channel members **2080**. Due to the greater thickness of the bottom sacrificial layer **206B**, the cavity **262** remains unfilled by the liner **266**. The liner **266** allows selective deposition of a bottom anti-reflective coating (BARC) layer **268** into the cavity **262** at block **134**.

Referring to FIGS. **1B** and **18**, method **100** includes a block **134** where a bottom anti-reflective coating (BARC) layer **268** is deposited to fill in a cavity **262** below the APT member **2080B**. In some embodiments, the BARC layer **268** may include polysulfones, polyureas, polyurea sulfones, polyacrylates, poly(vinyl pyridine), or a silicon-containing polymer. At block **134**, the BARC layer **268** may be deposited over the workpiece **200** using spin-on coating or FCVD. The deposited BARC layer **268** is then etched back until no BARC layer **268** is above the APT member **2080B**.

Referring to FIGS. **1B** and **19**, method **100** includes a block **136** where the liner **266** and the first dielectric layer **264** are etched back. With the BARC layer **268** protecting the underlying liner **266** and the first dielectric layer **264**, the liner **266** and the first dielectric layer **264** above the APT member **2080B** are etched back at block **136**. In some embodiments, the etch back of the liner **266** and the first dielectric layer **264** is performed using a selective wet etch process or a selective dry etch process. An example selective wet etch process may include use of a high-temperature sulfuric peroxide mixture (SPM, also known as a piranha solution, including sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2)) or phosphoric acid (H_3PO_4). An example selective dry etch process may include use of boron trichloride (BCl_3).

Referring to FIGS. **1B** and **20**, method **100** includes a block **138** where the BARC layer **268** and the remaining liner **266** are etched back. At block **138**, the BARC layer **268** is first removed by ashing or stripping. After the removal of the BARC layer **268**, the exposed liner **266** is selectively removed using a selective wet etch process, such as a wet etch process that uses buffered hydrofluoric acid (BHF) or diluted hydrofluoric acid (DHF). With the removal of the BARC layer **268** and the liner **266**, the cavity **262** under the APT member **2080B** once again becomes vacant, albeit lined by the first dielectric layer **264**.

Referring to FIGS. **1B** and **21**, method **100** includes a block **140** where a second dielectric layer **272** is deposited to seal the cavity **262** below the APT member **2080B**. In some embodiments, the second dielectric layer **272** is also formed of a high-k dielectric material, such as aluminum oxide, zirconium oxide, zirconium aluminum oxide, hafnium oxide, other metal oxide, or a combination thereof. In one embodiment, the second dielectric layer **272** may be formed of hafnium oxide. The second dielectric layer **272** may be deposited using CVD or ALD. In some implementations, the second dielectric layer **272** may be deposited to a thickness between about 2 nm and about 6 nm to seal the cavity **262**, but not fill it up. In addition to sealing up the cavity **262**, the second dielectric layer **272** is also conformally deposited over the helmet layer **228**, the center dielectric fin **219**, the separation dielectric fins **225**, surfaces of the channel members **2080**, and the exposed surfaces of the APT member **2080B**. After the cavities **262** are sealed off by the second dielectric layer **272**, each of them may have

13

a height (along the Z direction) between about 3 nm and about 25 nm. The cavities **262** may also be referred to as gaps or capsules.

Referring to FIGS. 1B and **22**, method **100** includes a block **142** where the deposited second dielectric layer **272** is etched back. At block **142**, the second dielectric layer **272** that is above the APT member **2080B** is selectively removed using a selective wet etch process or a selective dry etch process. An example selective wet etch process may include use of a high-temperature sulfuric peroxide mixture (SPM, also known as a piranha solution, including sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2)) or phosphoric acid (H_3PO_4). An example selective dry etch process may include use of boron trichloride (BCl_3). The selective removal prepares the channel members **2080** for the upcoming deposition of gate structures. As shown in FIG. **22**, the first dielectric layer **264** and the second dielectric layer **272** surrounds and defines the cavity **262**. Each of the cavities **262** is disposed between an APT member **2080B** and a base portion **212B**. Along the X direction, each of the cavities **262** is disposed between a separation dielectric fin **225** and a center dielectric fin **219**.

Referring to FIGS. 1B and **23**, method **100** includes a block **144** where a first gate structure **280-1** and a second gate structure **280-2** are formed to wrap around each of the channel members **2080**. Each of the first gate structure **280-1** and the second gate structure **280-2** includes an interfacial layer **282** on the channel members **2080** and the APT member **2080B**, a gate dielectric layer **284** over the interfacial layer **282**, and a gate electrode layer **286** over the gate dielectric layer **284**. In some embodiments, the interfacial layer **282** includes silicon oxide and may be formed as result of a pre-clean process. An example pre-clean process may include use of RCA SC-1 (ammonia, hydrogen peroxide and water) and/or RCA SC-2 (hydrochloric acid, hydrogen peroxide and water). The pre-clean process oxidizes the exposed surfaces of the channel members **2080** and the APT member **2080B** to form the interfacial layer **282**. The gate dielectric layer **284** is then deposited over the interfacial layer **282** using ALD, CVD, and/or other suitable methods. The gate dielectric layer **284** may include high-K dielectric materials. In one embodiment, the gate dielectric layer **284** may include hafnium oxide. Alternatively, the gate dielectric layer **284** may include other high-K dielectrics, such as titanium oxide (TiO_2), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta_2O_5), hafnium silicon oxide (HfSiO_4), zirconium oxide (ZrO_2), zirconium silicon oxide (ZrSiO_2), lanthanum oxide (La_2O_3), aluminum oxide (Al_2O_3), zirconium oxide (ZrO), yttrium oxide (Y_2O_3), SrTiO_3 (STO), BaTiO_3 (BTO), BaZrO , hafnium lanthanum oxide (HfLaO), lanthanum silicon oxide (LaSiO), aluminum silicon oxide (AlSiO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), $(\text{Ba,Sr})\text{TiO}_3$ (BST), silicon nitride (SiN), silicon oxynitride (SiON), combinations thereof, or other suitable material.

After the formation or deposition of the interfacial layer **282** and the gate dielectric layer **284**, the gate electrode layer **286** is deposited over the gate dielectric layer **284**. The gate electrode layer **286** may be a multi-layer structure that includes at least one work function layer and a metal fill layer. By way of example, the at least one work function layer may include titanium nitride (TiN), titanium aluminum (TiAl), titanium aluminum nitride (TiAlN), tantalum nitride (TaN), tantalum aluminum (TaAl), tantalum aluminum nitride (TaAlN), tantalum aluminum carbide (TaAlC), tantalum carbonitride (TaCN), or tantalum carbide (TaC). The metal fill layer may include aluminum (Al), tungsten (W),

14

nickel (Ni), titanium (Ti), ruthenium (Ru), cobalt (Co), platinum (Pt), tantalum silicon nitride (TaSiN), copper (Cu), other refractory metals, or other suitable metal materials or a combination thereof. In various embodiments, the gate electrode layer **286** may be formed by ALD, PVD, CVD, e-beam evaporation, or other suitable process. Although not explicitly shown in the figures, the first gate structure **280-1** and the second gate structure **280-2** are deposited as a joint gate structure and then etched back until the helmet layer **228** separates the joint gate structure into the first gate structure **280-1** and the second gate structure **280-2**. As shown in FIG. **23**, each of the first gate structure **280-1** and the second gate structure **280-2** wraps around each of the channel members **2080** vertically stacked over a base portion **212B**. It is noted that neither the first gate structure **280-1** nor the second gate structure **280-2** extends between the center dielectric fin **219** and the channel members **2080** as the channel members **2080** are in contact with the center dielectric fin **219**.

Referring to FIGS. 1B and **24**, method **100** includes a block **146** where the helmet layer **228** is etched back. In some embodiments represented in FIG. **24**, the helmet layer **228** may be selectively removed using a selective dry etch process or a selective wet etch process. An example selective wet etch process may include use of a high-temperature sulfuric peroxide mixture (SPM, also known as a piranha solution, including sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2)) or phosphoric acid (H_3PO_4). An example selective dry etch process may include use of boron trichloride (BCl_3). The etch back at block **146** does not etch the first gate structure **280-1** and the second gate structure **280-2** substantially or etches them at a slower rate than it etches the helmet layer **228**.

Referring to FIGS. 1B and **25**, method **100** includes a block **148** where a metal cap layer **288** is deposited. After the removal of the helmet layer **228**, the metal cap layer **288** is deposited over the first gate structure **280-1**, the second gate structure **280-2**, the center dielectric fin **219**, and the separation dielectric fins **225**. In some embodiments, the metal cap layer **288** may include titanium (Ti), titanium nitride (TiN), tantalum nitride (TaN), tungsten (W), ruthenium (Ru), cobalt (Co), or nickel (Ni) and may be deposited using PVD, CVD, or metal organic chemical vapor deposition (MOCVD). In one embodiment, the metal cap layer **288** includes tungsten (W) and is deposited by PVD. When not separated by a gate cut feature **292** (to be described below), the metal cap layer **288** electrically couples the first gate structure **280-1** and the second gate structure **280-2**.

Referring to FIGS. 1B and **26**, method **100** includes a block **150** where a self-aligned cap (SAC) layer **290** is formed over the gate structures. After the deposition of the metal cap layer **288**, the SAC layer **290** is deposited over the workpiece **200** by CVD, PECVD, or a suitable deposition process. The SAC layer **290** may include silicon oxide, silicon nitride, silicon carbide, silicon carbonitride, silicon oxynitride, silicon oxycarbonitride, aluminum oxide, aluminum nitride, aluminum oxynitride, zirconium oxide, zirconium nitride, zirconium aluminum oxide, hafnium oxide, or a suitable dielectric material. Photolithography processes and etch processes are then performed to etch the deposited SAC layer **290** to form a gate cut opening to expose the top surfaces of the center dielectric fin **219**.

Referring to FIGS. 1B and **26**, method **100** includes a block **152** where a gate cut feature **292** is formed over the center dielectric fin. Thereafter, a dielectric material is deposited and planarized by a CMP process to form the gate cut features **292** in the gate cut opening over the center

15

dielectric fin 219. The dielectric material for the gate cut features 292 may be deposited using HDPCVD, CVD, ALD, or a suitable deposition technique. In some instances, the gate cut feature 292 may include silicon oxide, silicon nitride, silicon carbide, silicon carbonitride, silicon oxynitride, silicon oxycarbonitride, aluminum oxide, aluminum nitride, aluminum oxynitride, zirconium oxide, zirconium nitride, zirconium aluminum oxide, hafnium oxide, or a suitable dielectric material. In some embodiments, the gate cut feature 292 and the SAC layer 290 may have different compositions to introduce etch selectivity. The portion of the first dielectric layer 264 and the second dielectric layer 272 that extends between the APT member 2080B and the separation dielectric fin 225 may be referred to as an end cap 294. As shown in FIG. 26, the end cap 294 may have a width W along the X direction. The width W may be between about 7 nm and about 18 nm. Additionally, another portion of the first dielectric layer 264 and the second dielectric layer 272 may extend between a base portion 212B and a separation dielectric fin 225 and may be referred to as base end cap 296. A width of the base end cap 296 may be similar to the end cap 294.

FIG. 27 illustrates a fragmentary cross-sectional view across source/drain regions of the workpiece 200 after operations at block 152 are concluded. In some embodiments, due to faceted growth of the source/drain features 256, a portion of the CESL 258 may extend downward into the space between a source/drain feature 256 and a separation dielectric fin 225. Due to the recess operations to form the source/drain recesses, the APT member 2080B in the source/drain region may be thinner than that in the channel regions under either the first gate structure 280-1 or the second gate structure 280-2, as shown in FIG. 26. Because the selective removal of the bottom sacrificial layer 206B and the cladding layer 220 takes place after the formation of the inner spacer features 254, a portion of the inner spacer feature 254 is disposed between the APT member 2080B and the separation dielectric fin 225. That is, in the source/drain region, the APT member 2080B does not engage any end caps that are formed of the first dielectric layer 264 or the second dielectric layer 272.

Reference is made to FIGS. 26 and 27. Upon conclusion of method 100, two MBC transistors that are divided by the center dielectric fin 219 are formed. One of them is controlled by the first gate structure 280-1 and the other is controlled by the second gate structure 280-2. In some embodiments, these two MBC transistors are of the same conductivity type or of different conductivity types. For example, both MBC transistors may be p-type with p-type source/drain features 256 and n-type APT members 2080B. For another example, both MBC transistors may be n-type with n-type source/drain features 256 and p-type APT members 2080B. Because the channel members 2080 are in contact with and stem from sidewalls of the center dielectric fin 219, the MBC transistors may be referred to as fork-sheet transistors or fish-bone transistors. Alternatively, these MBC transistors may be referred to as MBC transistors with fish-bone/fork-sheet structures.

Based on the above discussions, it can be seen that the present disclosure offers advantages. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. For example, the present disclosure provides fork-sheet transistors where channel members on either side of a center dielectric fin are disposed over an anti-punch-through (APT) member that stems from the center dielectric

16

fin. A cavity or capsule below the APT member further isolate the channel members from base portions that are connected to the bulk substrate. The APT members and the cavities below the APT members may reduce leakage through the bulk substrate.

In one aspect, the present disclosure provides embodiments of a semiconductor structure that includes a first base portion and a second base portion, an isolation feature sandwiched between the first base portion and the second base portion, a center dielectric fin over the isolation feature, a first anti-punch-through (APT) feature over the first base portion, a second APT feature over the second base portion, a first stack of channel members over the first APT feature, and a second stack of channel members over the second APT feature. The center dielectric fin is sandwiched between the first stack of channel members and the second stack of channel members as well as between the first APT feature and the second APT feature.

In some embodiments, the first stack of channel members are in contact with a source/drain feature and the source/drain feature and first APT feature are doped with different types of dopants. In some implementations, the semiconductor structure may further include a first cavity disposed between the first base portion and the first APT feature and a second cavity disposed between the second base portion and the second APT feature. The center dielectric fin extends between the first cavity and the second cavity. In some instances, the first cavity is defined in at least one dielectric layer. In some embodiments, the at least one dielectric layer includes aluminum oxide, zirconium oxide, zirconium aluminum oxide, hafnium oxide, or a combination thereof. In some embodiments, the semiconductor structure may further include a first gate structure wrapping around each of the first stack of channel members and a second gate structure wrapping around each of the second stack of channel members. The first gate structure does not extend between the first APT feature and the first base portion and the second gate structure does not extend between the second APT feature and the second base portion. In some instances, the first gate structure is sandwiched between the center dielectric fin and a separation dielectric fin. In some embodiments, the first APT feature is in contact with the center dielectric fin and spaced apart from the separation dielectric fin by the at least one dielectric layer. In some implementations, a portion of the at least one dielectric layer extends between the separation dielectric fin and the first base portion.

In another aspect, the present disclosure provides embodiments of a semiconductor structure that includes a first base portion and a second base portion, an isolation feature sandwiched between the first base portion and the second base portion, a center dielectric fin over the isolation feature, a first anti-punch-through (APT) feature over the first base portion, a second APT feature over the second base portion, a first source/drain feature over the first APT feature, and a second source/drain feature over the second APT feature. The center dielectric fin is sandwiched between the first source/drain feature and the second source/drain feature as well as between the first APT feature and the second APT feature.

In some embodiments, the first source/drain feature and the second source/drain feature are doped with a dopant of a first conductivity type and the first APT feature and the second APT feature are doped with a dopant of a second conductivity type different from the first conductivity type. In some implementations, the semiconductor structure may further include a first cavity disposed between the first base portion and the first APT feature and a second cavity

17

disposed between the second base portion and the second APT feature. The center dielectric fin extends between the first cavity and the second cavity. In some embodiments, the first cavity is defined in at least one dielectric layer. In some embodiments, the at least one dielectric layer includes aluminum oxide, zirconium oxide, zirconium aluminum oxide, hafnium oxide, or a combination thereof. In some instances, the first source/drain feature is sandwiched between the center dielectric fin and a separation dielectric fin. In some instances, the first APT feature is in contact with the center dielectric fin and spaced apart from the separation dielectric fin by an inner spacer feature. In some implementations, a portion of the at least one dielectric layer extends between the separation dielectric fin and the first base portion.

In yet another aspect, the present disclosure provides embodiments of a method that includes depositing a bottom sacrificial layer over a substrate, depositing a doped silicon layer over the bottom sacrificial layer, forming a stack over the doped silicon layer, the stack including a plurality of channel layers interleaved by a plurality of sacrificial layers, depositing a top sacrificial layer over the stack, patterning the top sacrificial layer, the stack, the doped silicon layer, the bottom sacrificial layer, and a portion of the substrate to form a first fin-shaped structure and a second fin-shaped structure, forming a center dielectric fin between the first fin-shaped structure and the second fin-shaped structure, forming a helmet layer on the center dielectric fin, and selectively removing the top sacrificial layer from the first fin-shaped structure and the second fin-shaped structure.

In some embodiments, the method may further include forming a dummy gate stack over channel regions of the first fin-shaped structure and the second fin-shaped structure, recessing source/drain regions of the first fin-shaped structure and the second fin-shaped structure, selectively recessing the plurality of sacrificial layers to form inner spacer features, forming a first source/drain feature in contact with the doped silicon layer, removing the dummy gate stack, selectively removing the sacrificial layers and the bottom sacrificial layer in the first fin-shaped structure and the second fin-shaped structure to release the plurality of channel layers as channel members and the doped silicon layer as an anti-punch-through (APT) feature, conformally depositing a first high-k dielectric layer over surfaces of the channel members and the APT feature, and conformally depositing an oxide liner over the first high-k dielectric layer. In some implementations, the method may further include depositing a bottom anti-reflective coating (BARC) layer over the oxide liner, after the depositing of the BARC layer, etching back the BARC until a top surface of the BARC layer is between a top surface of the APT feature, etching back the oxide liner and the first high-k dielectric layer, after the etching back of the oxide liner and the first high-k dielectric layer, selectively removing BARC layer to form a cavity below the APT feature, sealing the cavity with a second high-k dielectric layer, and after the sealing, forming a gate structure to wrap around each of the channel members.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may

18

make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:

- a substrate;
- a first base portion and a second base portion over the substrate;
- an isolation feature sandwiched between the first base portion and the second base portion;
- a center dielectric fin over the isolation feature and comprising a first sidewall and a second sidewall opposing the first sidewall;
- a first source/drain feature disposed over the first base portion and extending along the first sidewall of the center dielectric fin;
- a second source/drain feature disposed over the second base portion and extending along the second sidewall of the center dielectric fin;
- a first cavity disposed between the first source/drain feature and the first base portion; and
- a second cavity disposed between the second source/drain feature and the second base portion.

2. The semiconductor structure of claim 1, further comprising:

- a first anti-punch-through (APT) feature disposed between the first source/drain feature and the first cavity; and
- a second APT feature disposed between the second source/drain feature and the second cavity.

3. The semiconductor structure of claim 2, wherein the first cavity is defined in a first dielectric layer below a top surface of the first APT feature, wherein the second cavity is defined in a second dielectric layer below a top surface of the second APT feature.

4. The semiconductor structure of claim 3, wherein the first APT feature is partially disposed in the first dielectric layer, wherein the second APT feature is partially disposed in the second dielectric layer.

5. The semiconductor structure of claim 2, wherein the first APT feature and the second APT feature comprise silicon and a dopant, wherein the dopant comprises phosphorus (P), arsenic (As), or boron (B).

6. The semiconductor structure of claim 2, wherein sidewalls of the first source/drain feature and the first APT feature interface the first sidewall, wherein sidewalls of the second source/drain feature and the second APT feature interface the second sidewall.

7. The semiconductor structure of claim 1, wherein the center dielectric fin comprises an outer layer in contact with the first source/drain feature, the second source/drain feature, and the isolation feature and an inner layer spaced apart from the first source/drain feature, the second source/drain feature, and the isolation feature.

8. The semiconductor structure of claim 7, wherein the outer layer comprises silicon nitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, aluminum nitride, aluminum oxynitride, zirconium nitride, or silicon oxynitride, wherein the inner layer comprises silicon oxide.

9. The semiconductor structure of claim 1, further comprising: a contact etch stop layer (CESL) extending continuously from over a top surface of the first source/drain feature,

19

a top surface of the center dielectric fin, and a top surface of the second source/drain feature.

10. A semiconductor structure, comprising:

a substrate;

a first base portion and a second base portion over the substrate;

an isolation feature sandwiched between the first base portion and the second base portion;

a center dielectric fin over the isolation feature and comprising a first sidewall and a second sidewall opposing the first sidewall;

a first source/drain feature disposed over the first base portion and extending along the first sidewall of the center dielectric fin;

a second source/drain feature disposed over the second base portion and extending along the second sidewall of the center dielectric fin;

a first dielectric layer disposed between the first source/drain feature and the first base portion; and

a second dielectric layer disposed between the first source/drain feature and the first base portion, wherein the first dielectric layer comprises a first cavity, wherein the second dielectric layer comprises a second cavity.

11. The semiconductor structure of claim **10**, wherein the first dielectric layer and the second dielectric layer comprise aluminum oxide, zirconium oxide, zirconium aluminum oxide, hafnium oxide, or other metal oxide.

12. The semiconductor structure of claim **10**, further comprising

a first dielectric fin adjacent the first source/drain feature; and

a second dielectric fin adjacent the second source/drain feature,

wherein the first source/drain feature is sandwiched between the first dielectric fin and the center dielectric fin along a direction,

wherein the second source/drain feature is sandwiched between the second dielectric fin and the center dielectric fin along the direction.

13. The semiconductor structure of claim **12**,

wherein the first dielectric layer extends between the first dielectric fin and the center dielectric fin along the direction,

wherein the second dielectric layer extends between the second dielectric fin and the center dielectric fin along the direction.

14. The semiconductor structure of claim **10**, further comprising:

a first anti-punch-through (APT) feature disposed between the first source/drain feature and the first dielectric layer; and

a second APT feature disposed between the second source/drain feature and the second dielectric layer.

15. The semiconductor structure of claim **14**,

wherein the first APT feature and the second APT feature comprise silicon and a dopant,

wherein the dopant comprises phosphorus (P), arsenic (As), or boron (B).

20

16. The semiconductor structure of claim **14**,

wherein sidewalls of the first source/drain feature, the first APT feature and the first dielectric layer interface the first sidewall,

wherein sidewalls of the second source/drain feature, the second APT feature, and the second dielectric layer interface the second sidewall.

17. A semiconductor structure, comprising:

a substrate;

a first base portion and a second base portion over the substrate;

an isolation feature sandwiched between the first base portion and the second base portion along a direction;

a center dielectric fin over the isolation feature and comprising a first sidewall and a second sidewall opposing the first sidewall;

a first source/drain feature disposed over the first base portion and extending along the first sidewall of the center dielectric fin;

a second source/drain feature disposed over the second base portion and extending along the second sidewall of the center dielectric fin;

a first dielectric layer disposed between the first source/drain feature and the first base portion; and

a second dielectric layer disposed between the first source/drain feature and the first base portion, wherein the first dielectric layer comprises a first cavity, wherein the second dielectric layer comprises a second cavity,

wherein a width of the center dielectric fin along the direction is the same as a width of the isolation feature along the direction.

18. The semiconductor structure of claim **17**, further comprising

a first dielectric fin adjacent the first source/drain feature; and

a second dielectric fin adjacent the second source/drain feature,

wherein the first source/drain feature is sandwiched between the first dielectric fin and the center dielectric fin along the direction,

wherein the second source/drain feature is sandwiched between the second dielectric fin and the center dielectric fin along the direction.

19. The semiconductor structure of claim **18**, further comprising:

a contact etch stop layer (CESL) extending continuously from over a top surface of the first dielectric fin, a top surface of the first source/drain feature, a top surface of the center dielectric fin, a top surface of the second source/drain feature, and a top surface of the second dielectric fin.

20. The semiconductor structure of claim **18**,

wherein the center dielectric fin comprises an outer layer in contact with the first source/drain feature, the second source/drain feature, and the isolation feature and an inner layer spaced apart from the first source/drain feature, the second source/drain feature, and the isolation feature.

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